Supplementary Materials for Evolutionary history from genomes to single cells by lossless compression

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Materials and Methods

Two-Part Minimum Description Length Objective and Star-Tree Null Baseline

In probabilistic phylogenetics, a substitution model and tree T assign a joint likelihood $P(A \mid T)$ to a discrete character matrix or multiple sequence alignment $A \in \Sigma^{N\times L}$ over N taxa (or single cells) and L columns on alphabet Σ (*2*). By Shannon's source coding theorem, $-\ln P(A \mid T)$ equals the exact code length in nats (1 nat = $1/\ln 2 \approx 1.4427$ bits) required to transmit A conditional on T. Standard model-selection criteria such as AIC and BIC (*50–53*) penalize the 1 to 10 continuous parameters of a rate matrix Q while leaving the $(N-3)$ discrete bipartition choices of T unpriced (*54, 55*). Under Rissanen's two-part Minimum Description Length (MDL) principle (*7, 8*), both the evolutionary graph $\mathcal{G} = (V, E_{\text{tree}}, E_{\text{shunt}})$ and the data encoded along it share one unit: $\mathcal{L}_{\text{MDL}}(A, \mathcal{G}) = \mathcal{L}(\mathcal{G}) + \mathcal{L}(A \mid \mathcal{G})$.

To establish an assumption-free null baseline, we price A on an unresolved star phylogeny T_{star} containing zero internal splits. Sending raw ungapped entries independently costs $NL_{\text{eff}} \ln|\Sigma|$ nats, whereas sending one ancestral root vector $X_r \in \Sigma^L$ costs $L \ln|\Sigma|$ nats plus, on each terminal branch $i \in \{1, \ldots, N\}$, only the m_i positions where lineage i differs across its $L_{\text{eff},i}$ observed columns:

$$\mathcal{L}_{\text{branch}}(m_i, L_{\text{eff},i}) \;=\; \mathcal{L}_{\mathbb{N}}(m_i) + \ln\binom{L_{\text{eff},i}}{m_i} + \sum_{k=1}^{m_i} \ell_{\text{subst}}(a_{i,k} \to b_{i,k}), \tag{1}$$

where $\mathcal{L}_{\mathbb{N}}(0) = \ln 2$ and, for $m_i \geq 1$,

$$\mathcal{L}_{\mathbb{N}}(m_i) \;=\; -\ln\left[\frac{\ln 2}{2}\left(\frac{1}{\ln(m_i+1)} - \frac{1}{\ln(m_i+2)}\right)\right]$$

is the Kraft-tight Elias–Rissanen universal integer code (*8*) satisfying $\sum_{m=0}^{\infty} e^{-\mathcal{L}_{\mathbb{N}}(m)} = 1$ identically without any separate active-branch indicator bit; $\ln \binom{L_{\text{eff},i}}{m_i}$ indexes which unordered subset of m_i columns mutated (vanishing at $m_i = 0$); and $\ell_{\text{subst}}(a \to b) = -\ln P(b \mid a)$ is the Kraft-normalized substitution surprise ($\sum_{b \neq a} e^{-\ell_{\text{subst}}(a \to b)} = 1$). Summing Eq. (1) over X_r and all leaves yields $\mathcal{L}_{\text{MDL}}(A, T_{\text{star}})$ and the dimensionless Phylogenetic Compression Ratio:

$$\rho(A) \;=\; 1 - \frac{\mathcal{L}_{\text{MDL}}(A, \mathcal{G}^*)}{\mathcal{L}_{\text{MDL}}(A, T_{\text{star}})}. \tag{2}$$

Exact Bipartition Cost, Rate Weighting, and the Natural Polytomy Law

Splitting a multifurcating hub w of degree $d = k_u + k_v \geq 4$ by inserting branch $e = (u, v)$ partitions its d incident lineages into $2 \leq k_u \leq k_v$ children. Indexing the unordered split size $\{k_u, k_v\}$ among $\max(1, \lfloor d/2 \rfloor - 1)$ non-trivial choices and selecting the k_u-subset (halved by symmetry when $k_u = k_v$) incurs the exact topological address cost:

$$\mathcal{L}_{\text{topo}}(k_u, k_v) \;=\; \ln \max\left(1,\; \left\lfloor \frac{k_u + k_v}{2} \right\rfloor - 1\right) \;+\; \ln \binom{k_u + k_v}{k_u} \;-\; \mathbb{I}(k_u = k_v) \ln 2 \quad \text{nats.} \tag{3}$$

For a quartet ($k_u = k_v = 2$), $\mathcal{L}_{\text{topo}}(2, 2) = \ln 3 \approx 1.099$ nats; within a degree-d burst hub, $\mathcal{L}_{\text{topo}}(2, d - 2) \approx 3 \ln d - 2 \ln 2$ automatically corrects for searching across all $\binom{d}{2}$ pairs (unlike purely local edge tests (*56, 57*)). Shifting one shared derived edit off an adjacent branch c carrying m_c mutations reduces its positional subset cost by

$$w_c \;=\; \ln \binom{L_{\text{eff}}}{m_c + 1} - \ln \binom{L_{\text{eff}}}{m_c} \;=\; \ln\left(\frac{L_{\text{eff}} - m_c}{m_c + 1}\right) \;\approx\; -\ln\left(\frac{p_c}{1 - p_c}\right) \quad \text{nats,} \tag{4}$$

recovering Felsenstein's exact maximum-likelihood log-odds branch weight (*2*). Writing $\Delta\mathcal{L}_{\text{children}}(m)$ for the child-branch code savings when m shared edits are factored onto e, split e is retained if and only if its net MDL compression gain is positive:

$$G_{\text{MDL}}(e) \;\equiv\; -\Delta\mathcal{L}_{\text{MDL}}(e) \;=\; \max_{1 \leq m \leq K} \left[\Delta\mathcal{L}_{\text{children}}(m) - \mathcal{L}_{\text{branch}}(m, L_{\text{eff}})\right] - \mathcal{L}_{\text{topo}}(k_u, k_v) \;>\; 0, \tag{5}$$

which maps directly onto a conditional local posterior branch probability $P_{\text{local}}(e \mid T_{-e}, A) = \left(1 + \exp(-G_{\text{MDL}}(e))\right)^{-1}$ comparing the locally optimal split state against collapsing e on the fixed surrounding topology T_{-e} without bootstrap resampling (*9*).

Four-Stage Hybrid Compression and Graph Induction Architecture

Finding the global MDL minimum across all $(2N - 5)!!$ leaf-labeled topologies is NP-hard, so Bushwhacker executes a deterministic four-stage search combining sub-quadratic distance/Voronoi seeding, vectorized continuous-time (`GTR+CAT8/LG08+CAT8`) or discrete/ordered (`-discrete`/`-ordered`) local optimization, post-hoc MDL contraction, and lateral shunt peeling:

1. **Bit-Parallel Encoding and Empirical Exchange Costs:** Nucleotide states and IUPAC ambiguities pack into four 64-bit integer bitplanes (`[u64; 4]`) per 64 alignment columns; variable-site patterns deduplicate deterministically by run-length encoding. Empirical transition ($\hat{p}_{\text{ti}}$) and transversion ($\hat{p}_{\text{tv}} = 1 - \hat{p}_{\text{ti}}$) costs ($\ell_{\text{ti}} = -\ln \hat{p}_{\text{ti}}$, $\ell_{\text{tv}} = -\ln(\hat{p}_{\text{tv}}/2)$) and empirical GTR/LG08 Markov parameters parameterize the discrete ledger and continuous-time likelihood sweeps.

2. **Stage 1 — Sub-Quadratic Starting Scaffold (T_0):** On a stratified pilot subsample ($K_{\text{sub}} \le 128$), sitewise Fitch edit counts s_k (*1*) under Poisson saturation $m_k = -k_{\text{sat}} \ln(1 - \min(0.92, s_k/k_{\text{sat}}))$ estimate the Gamma rate-dispersion shape $\hat{\alpha} \in [0.15, 50.0]$ and Jin–Nei (K80 + $\Gamma(\hat{\alpha})$) distances (*58*):

$$\begin{aligned} \hat{\alpha} &= \frac{\bar{m}^2}{\max\left(10^{-6},\ \mathrm{Var}(m) - \bar{m}\right)}, \\ M_{ij} &= L_{ij} \cdot \frac{\hat{\alpha}}{2}\left[(1 - 2P_{ij} - Q_{ij})^{-1/\hat{\alpha}} + \frac{1}{2}(1 - 2Q_{ij})^{-1/\hat{\alpha}} - \frac{3}{2}\right]. \end{aligned} \tag{6}$$

   For $N \le 4{,}096$, BioNJ builds the binary starting scaffold T_0 from the distance matrix; for $N > 4{,}096$, Voronoi landmark divide-and-conquer partitions taxa and builds an $O(N \log N \cdot L/64)$ skeleton (*59*).

3. **Stage 2 — Continuous-Time Rate-Category and Discrete Topology Refinement:** On all default nucleotide ($N \ge 4$) and protein/translated ($N \ge 3$) alignments, Bushwhacker computes the global empirical substitution density $\bar{\mu} = \sum_s s_k / \sum_s n_v(s)$ across active columns on the Stage 1 distance scaffold T_0 (where s_k is the column Fitch parsimony score across $n_v(s)$ non-missing taxa), estimates the shrunk site relative rate $\hat{r}_s = \mathrm{clamp}\big((\alpha_0 + s_k)/(\alpha_0 + \bar{\mu}\, n_v(s)),\ [r_{\min}, r_{\max}]\big)$ with $\alpha_0 = 1.5$ and $[r_{\min}, r_{\max}] = [0.065, 3.10]$ under fixed $\Gamma(1,1)$ octile rate categories $(r_1, \ldots, r_8,\ \frac{1}{8}\sum_{k=1}^{8} r_k = 1$; or $\alpha_0 = \hat{\alpha}$ and $[0.5r_1, 1.5r_8]$ under `--empirical-alpha`), assigns each variable column s unconditionally to $k^*(s) = \arg\min_k |\ln \hat{r}_s - \ln r_k|$, and runs a fixed deterministic schedule (`NNI + LeafMLE + SPR2 + NNI`) over 64-pattern SoA cache tiles (`GTR+CAT8` or `LG08+CAT8`): local `NNI` swaps accept Fisher-curvature-penalized log-likelihood gains $\Delta\text{MDL} > 0.01$ nats, while 15-topology `SPR2` quintet moves accept log-likelihood gains $\Delta \ln P > 0.05$ nats to prevent numerical oscillation. Under `-discrete` and `-ordered`, Stage 2 instead optimizes the exact discrete/ordered MDL objective directly via postorder/preorder Sankoff–Fitch/interval sweeps.

4. **Stage 3 — Deterministic Polytomy Contraction and Discrete Hub Expansion:** Internal branches contract sequentially from most negative gain upward when $G_{\text{MDL}}(e) \le 0$. On default continuous-time `GTR+CAT8/LG08+CAT8` paths, branch contraction tests $\Delta \ln P(e) - \left[\frac{1}{2}\ln \max(1, m_{\text{var}}/2\pi) + \frac{1}{2}\ln \max(1, N-3) + \mathcal{L}_{\text{topo}}(k_u, k_v)\right] \le 0$ and emits the contracted graph (alongside the pre-contraction binary refinement via `-b`) without running multi-taxon clique expansion; on `-discrete` and `-ordered` inputs, degree-d hubs additionally propose multi-taxon splits ($2 \le k_u \le d - 2$) retained whenever $G_{\text{MDL}}(e) > 0$.

5. **Stage 4 — Dual-Mode Lateral Shunt Peeling:** Residual non-sister parallel substitutions $S_{u,v} = \{s_1, \ldots, s_m\}$ (pooled across up to three consecutive ancestral coding branches while excluding ancestral symplesiomorphies) are tested as lateral links $h = (u \to v)$ priced by endpoint address $2 \ln |V|$ plus the shorter of contiguous-tract (`INTERVAL`) or scattered-site (`SPARSE`) mask codes:

$$\begin{aligned} \mathcal{L}_{\text{mask}}^{\text{INTERVAL}}([a,b]) &= \ln 2 + 2\ln L + \mathcal{L}_{\text{branch}}(d_{[a,b]}, b - a + 1), \\ \mathcal{L}_{\text{mask}}^{\text{SPARSE}}(S_{u,v}) &= \ln 2 + \ln \binom{M_{\text{pool}}}{m} + \mathcal{L}_{\mathbb{N}}(m - m_{\min}), \end{aligned} \tag{7}$$

   where $d_{[a,b]}$ counts discordant positions in $[a, b]$, $M_{\text{pool}} \le L$ is the size of the address space from which the m parallel sites are drawn—the number of variable alignment columns for

non-coding nucleotide, protein, and somatic alignments, and the number of amino-acid-variable codons (positions with ≥ 2 distinct residues; 302 of 745 codons in mammalian *Prestin*) for coding/translated alignments, matching the codon-level unit at which parallel coding substitutions are enumerated—and $m_{\min} = 2$ (nucleotide) or 4 (protein) is the structural minimum site threshold with Elias-delta / universal integer prefix code $\mathcal{L}_{\mathbb{N}}$ ensuring Kraft-exact self-delimiting completeness across shunt cardinalities (empirically producing 0/50 new shunts on both 20-state AA and nucleotide zero-shunt TOGA2 controls; note that setting M_{pool} to variable codons reflects causal coding enumeration rather than nuisance-parameter invariance: substituting full alignment length $L_{\text{codon}} = 745$ adds 3.624 nats to $\mathcal{L}_{\text{mask}}^{\text{SPARSE}}$ and lowers the 116-mammal *Prestin* sonar link from +3.912 nats [$P_{\text{local}} = 98.04\%$] to +0.288 nats [$P_{\text{local}} = 57.16\%$]). To guard against tearing genuine vertical tree splits on noisy tracts, Stage 4 masks recipient nucleotides only for contiguous `INTERVAL` transfers whose MDL gain exceeds 15 nats ($e^{-15} \approx 3 \times 10^{-7}$ false-masking odds) before running Stage 3 re-verification.

Multi-Alphabet Generalization and Continuous-Time Rate-Category Cache Tiling

Bushwhacker extends Eqs. (1)–(7) natively across discrete alphabets:

- **20-State Amino-Acid Alignments (`-protein` / `-translate`):** Twenty amino-acid bitplanes (`[u64; 20]` per 64 columns) and six Dayhoff–Miyata biochemical classes (*60*) yield empirical conservative probability $\hat{p}_{\text{cons}}$ (estimated from pairs with divergence ≤ 0.35) and normalized substitution costs:

$$\ell_{\text{subst}}^{\text{AA}}(a \to b) = \begin{cases} -\ln\big(\hat{p}_{\text{cons}}/k_{\text{same}}(a)\big), & \text{if class}(b) = \text{class}(a), \\ -\ln\big((1-\hat{p}_{\text{cons}})/k_{\text{diff}}(a)\big), & \text{if class}(b) \neq \text{class}(a). \end{cases} \tag{8}$$

- **Discrete Single-Cell Somatic Matrices (`-discrete`, $2 \leq K \leq 32$):** Binary somatic SNV profiles (0 = WT, 1 = mutant, ?/- = allelic dropout) assign full wildcard bitmasks ($2^K - 1$) to missing entries, incurring zero dropout penalty while adjusting $L_{\text{eff},i}$ per cell.

- **Ordered Zygosity and Copy-Number Profiles (`-ordered`, $3 \leq K \leq 16$):** Somatic zygosity states (0 = WT, 1 = HET, 2 = HOM/CN-LOH (*27, 47*)) and DLP+ copy-number bins ($0, 1, \ldots, 11+$ (*48*)) replace unordered Fitch intersections with exact linear-time L_1 interval propagation: parent interval $[l_1, r_1] \cap [l_2, r_2]$ if overlapping, or $[\min(r_1, r_2), \max(l_1, l_2)]$ otherwise, paired with geometric step costs $\ell_{\text{step}}(\Delta s) = -\ln\big((1-\hat{q})\hat{q}^{|\Delta s|-1}/Z_a\big)$. On 500-kb bin matrices (4,375 bins), Bushwhacker supports raw bin compression with root run-length chromosomal coding and lossless breakpoint-segment aggregation (365–448 multiplicity-weighted segments).

- **Continuous-Time Hard-CAT Rate Models with 64-Site Cache Tiling (`GTR+CAT8` / `LG08+CAT8`):** Running unconditionally on nucleotide ($N \geq 4$) and protein/translated ($N \geq 3$) alignments (*15, 16, 61*), continuous-time reversible Markov models assign each variable column s unconditionally by nearest log-rate to one of eight fixed categories ($r_1, \ldots, r_8$; analytical octile

means of $\Gamma(1,1)$ satisfying $\frac{1}{8}\sum_{k=1}^{8} r_k = 1$):

$$G_{\text{CT-MDL}}(e) = \Delta \ln P(e \mid T, r_{k^*(\cdot)}\hat{\mathbf{t}}, Q_M) - \left[\frac{1}{2}\ln \max\left(1, \frac{m_{\text{var}}}{2\pi}\right) + \frac{1}{2}\ln \max(1, N-3) + \mathcal{L}_{\text{topo}}(k_u, k_v)\right], \quad (9)$$

evaluated over cached transition matrices $P(t) = U\,\text{diag}(e^{\lambda r_k t})U^{-1}$ on a 15-point log-spaced branch-length grid with parabolic curvature refinement ($\frac{1}{2}\ln(1+\mathcal{I}(e)/2\pi)$ during NNI scoring), contracting silent stems smoothly when $G_{\text{CT-MDL}}(e) \le 0$. Partitioning unique site patterns into $B = 64$-pattern L1/L2 cache tiles bounds resident likelihood memory at $O(N \cdot B \cdot C \cdot |\Sigma|)$ (1.98–2.71 GB across mammalian and prokaryotic marker corpora, and 4.07 GB at 32,768 taxa vs. 26.37–79.15 GB for standard likelihood engines; Table S11). Controlled four-mode ablations (`-no-mult-pen` omitting $\frac{1}{2}\ln\max(1, N-3)$ and `-empirical-alpha` estimating discrete α in place of $\Gamma(1,1)$; Section S2 and Table S11) verify that both architectural choices remain empirically necessary and non-fragile. In JSON output, `tree_MDL_nats` records the unified discrete branch-edit summary code (Eq. (1) evaluated at continuous-time-informed branch loads $\max(\hat{t}_e \bar{r}_{\text{emp}} L_{\text{eff}}, m_e)$), whereas topological contraction on nucleotide and protein alignments evaluates the continuous-time criterion of Eq. (9) directly.

Lossless `.bwz` Causal Phylogenetic Arithmetic Codec

Invoked via `bushwhacker -c` and `-d`, the `.bwz` archiver encodes taxon identifiers, the Newick graph, the root sequence (with Markov run-length coding for indel and dropout runs), and causal preorder branch deltas using 32-bit arithmetic coding (*62*) conditioned on empirical site-rate and codon-phase priors, guaranteeing bit-exact FASTA reconstruction (`SHA-256` verified across Table S9).

Computational Environment, Baseline Software, and Multiple-Testing Governance

- **Implementation and Determinism:** Bushwhacker is written in native Rust (`rustc 1.94.1`) with zero external C/C++ dependencies and compiles to both a multi-threaded CLI binary (`-T`/`-threads`, bit-exact identical across 1–64 threads with fixed seed `20261112` and zero random tie-breaking) and client-side WebAssembly (`bushwhacker_bg.wasm`). Open-source code, specimen files, reproducibility scripts, and browser UI reside at `https://github.com/veg/bushwhacker` and `https://veg.github.io/bushwhacker/`.

- **Compute Hardware:** All 32-core Tree-of-Life supermatrices, Greengenes 16S scaling runs, pandemic viral alignments, and coalescent cluster sweeps ran on an ARM64 Neoverse-N1 Linux server (64 cores, pinning 32 threads per method in head-to-head comparisons unless noted, with identical hardware wall-clock quotas and no timeouts bypassed). Single-cell cancer phylogenomics, specimen reconstructions, and `.bwz` archive verifications ran on an Apple M5 Max workstation (18-core CPU, 36 GB unified memory, macOS).

- **External Baseline Software:** Baselines were executed with `IQ-TREE multicore v2.4.0` (`-m MFP`, `-m GTR+I+G4`, `-m GTR+G4 -fast`, `-m LG+G4 -fast`, `-polytomy`, `-abayes`, `-alrt 1000`, `-B 1000`, and `AliSim`), `FastTree/FastTreeMP v2.1.11` (`-nt -gtr`, `-lg`),

ASTRAL-IV (ASTER astral4 -t 8), GCtree, COMPASS (*47, 63*), HyPhy GARD, INDELible, gzip -9, bzip2 v1.0.8 -9, and xz/lzma (XZ Utils v5.8.1, -6/-9), scored by exact bipartition Robinson–Foulds matching (*64*).

- **Zero-Leakage, Multiple-Testing, and Ethics Governance:** Across every immune and cancer single-cell benchmark, cell multiplicities, DMS affinity scores, clinical timepoints, anatomic compartments, and passage numbers were stripped strictly prior to inference and linked solely post hoc. To control the false discovery rate (FDR) across all 31 two-sided biological hypothesis tests audited across main text and supplement (including ecological guild, archaeal ar122, and negative-control screens), raw p-values were adjusted jointly using Benjamini–Hochberg (BH) step-up control ($\alpha_{\mathrm{FDR}} = 0.05$): 24/31 tests remain significant at $q < 0.05$, whereas all seven null/negative-control comparisons remain non-significant ($q_{\mathrm{BH}} \geq 0.194$: non-multicopy ant-eating screen $p = 0.157, q = 0.194$; unadjusted marine/aquatic $p = 0.164, q = 0.196$; ar122 Complexity Hypothesis $p = 0.213, q = 0.221$; subterranean blind $p = 0.214, q = 0.221$; pooled binary BCR hub-vs-nonhub DMS affinity $p = 0.203, q = 0.221$; Morita AML direct cell count vs. k_{out} $p = 0.197, q = 0.221$; and DLP+ SA535 passage aneuploidy $p = 0.475, q = 0.475$). This retrospective *in silico* study analyzed only public de-identified datasets with zero wet-lab manipulation, human identifiers, or biosecurity concerns.

- **AI-Assisted Software Engineering and Manuscript Preparation:** Google Gemini large language models (Gemini 4 and Gemini 3.8 Flash, accessed via Google Antigravity) were used under author direction to assist in implementing and refactoring Rust/WebAssembly modules, browser visualization and Python plotting scripts, and manuscript copyediting. No generative AI tools were used to synthesize scientific figures or fabricate data; every mathematical derivation, algorithmic implementation, executed benchmark output, statistical test, and bibliographic reference was inspected, run, and verified by the human authors, who assume full responsibility for the work.

Data Provenance and Empirical Archive Identifiers

The complete self-contained Bushwhacker v1.0 replication archive (source code, specimens, empirical and simulated benchmark ledgers, single-cell cohorts, figure generators, and SHA-256 checksum manifest) is permanently archived at Zenodo (https://doi.org/10.5281/zenodo.24222789; preprint at https://doi.org/10.5281/zenodo.24222787) alongside https://github.com/veg/bushwhacker.

1. **Model-Selection Benchmark Archive (**$7,200$ **empirical +** $27,500$ **simulated alignments):** OSF repository https://doi.org/10.17605/OSF.IO/T3PF2 (*13, 65*), spanning PlantDB, Selectome, TreeBASE, and Pandora empirical loci alongside INDELible ground-truth suites sim_bic ($n = 5,901$), c0 ($n = 7,199$), c1 ($n = 7,200$), and c2 ($n = 7,200$).

2. **Eukaryotic Phylogenomic Radiations (**19 **datasets,** $45,205$ **loci;** $83,180$ **multi-seed searches):** Figshare project 22040 (https://figshare.com/projects/Evaluating_fast_maximum_likelihood-based_phylogenetic_programs_using_empirical_phylogenomic_data_sets/22040 (*12, 66*)), including mammalian CHST13 (22 species, $1,176$ bp from TarvD7), plus the whole-genome avian radiation (Avian_Shultz2019;

Dryad https://doi.org/10.5061/dryad.4b8gt83 (*38*); 50 clean coding orthologs evaluated in ASTRAL-IV alongside 11 radiation benchmarks ChenA4, MisoA2, ShenA9, BoroA6, WickA3/WickD3a, NagyA1, SongD1, XiD4, PrumD6, WhelA7 totaling 12,008 runs / 11,164 distinct loci).

3. **Mammalian *Prestin* (SLC26A5) and TOGA2 882-Mammal Atlas:** Complete 116-mammal *Prestin* coding alignment (2,235 bp; docs/examples/prestin.fasta (*19, 67*)), and the Senckenberg TOGA2 hg38 v1 archive (https://genome.senckenberg.de/download/TOGA2/MultiCodonAlignments/reference_human_hg38/v1/ (*23, 68*); 882 assemblies across 20 orders and 116 families: 19,952 coding alignments, 19,953 translated proteins, 5,058 de novo WALRUS codon MSAs with 1,487 completed protein-shunt cross-checks, and the 253-locus concatenated mammalian supermatrix).

4. **Botanical and Poaceae C_4 Corpus (2,002 alignments):** Land-plant WickD3a/WickD3b (1,688 alignments (*32*)), seed-plant XiD4 (310 genes (*69*)), GenBank plastid rbcL_483 (483 × 1,398 bp), and Poaceae ppc, pck, and rbcL orthologs/paralogs including *Alloteropsis semialata* C_3/C_4 copies retrieved from NCBI Entrez Nucleotide (*40, 70, 71*).

5. **Prokaryotic Domains, Clinical *K. pneumoniae*, and Greengenes 16S:** GTDB Release 95.0 (https://data.gtdb.ecogenomic.org/releases/release95/95.0/ (*25, 72*): ar122 across 1,672 archaea and 1,000-species bac120 across all 111 phyla); clinical *Klebsiella pneumoniae* genomes from NCBI BioProject PRJNA246471 and RefSeq (*42*) (16-strain 214,038-bp core chromosome supermatrix, 222 single-copy core loci across 10 genomes, and matched 9 clinical isolates—CAV1016, CAV1042, CAV1193, CAV1344, CAV1392, CAV1453, CAV1596, HS11286_ST11, NJST258_2—pairing the 214,038-bp chromosome against their 882-bp *blaKPC* plasmid); *S. pneumoniae* PMEN1 (11×2.22 Mb (*73*)); and Greengenes 13_5 16S rRNA (https://greengenes.secondgenome.com/ (*44*), $N = 256 \rightarrow 32,768 \times 1,348$ bp).

6. **Pandemic Viral and .bwz Archives:** Open Nextstrain/GenBank SARS-CoV-2 (https://data.nextstrain.org/files/ncov/open/; 5,097 and 10,062 complete 29,903-bp genomes plus blinded Pango X* recombinants), global Dengue DENV-1..4 (1,426, 2,632, 855, and 585 coding genomes; 5,498 total), LANL HIV-1 (https://www.hiv.lanl.gov/; KAL153, CRF02_AG, HIV1_518 (*74, 75*)), bat sarbecoviruses (Sarbeco_83, 83 × 30,965 bp (*41, 76*)), InfluenzaA HA (349 × 987 bp), Human_mtDNA (78 × 16,585 bp), hominoid MHC (379 × 1,128 bp (*77*)), and *Heliconius* optix (*78, 79*).

7. **Single-Cell Immune and Cancer Cohorts:** Mouse CGG-immunized germinal-center repertoires from GCReplay (https://github.com/matsengrp/gcreplay (*26*); 147 GCs, 10,429 paired IGH+IGK cells, 9,352 unique 657-bp genotypes, plus 4,400 scFv yeast-display $\Delta \log_{10} K_D$ scores) and Clone_3-8 (66 cells (*45*)); Mission Bio Tapestri myeloid scDNA runs from COMPASS (Zenodo https://doi.org/10.5281/zenodo.10822292 (*47*); 168 runs, 1,001,609 cells) and the 123-patient AML cohort including specimen AML-38-001 (7,016 cells, 11 clones ×31 loci, plus 7,016 × 16 raw single-cell Tapestri comparison against COMPASS; NCBI GEO GSE146150 and Zenodo https://zenodo.org/records/15096570 (*27*)); and DLP+ scWGS copy-number matrices (https://doi.org/10.24072/pcjournal.292 (*48*); 4,950 cells across OV2295, SA501, SA609, SA535).

8. **Ground-Truth Synthetic Benchmarks (36,920 + 100 GC histories):** `AliSim` 22-regime nucleotide suite (220 alignments (*80*)), `Abadi/Gerth` nucleotide suite (27,500 alignments), `LG+G4` protein suite (`P1-P4`, 400 alignments), deterministic Kingman Anomaly-Zone coalescent suite (`seed = 20261112`) (8,000 loci (*81*)) + 50 Galton–Watson affinity-selected B-cell GCs + 50 neutral ($\beta_{\text{bind}} = 0$) negative-control GCs (`MK_RS5NF` 5-mer SHM (*46*)), reticulate HGT/convergence suite (`N1-N4`, 400 alignments), and single-cell SNV/CNA matrices (`C1-C4`, 400 matrices).

Supplementary Text

Section S1: Why Two-Part Compression Resists Long-Branch Attraction and Enforces Polytomies

Counting edits along branches superficially resembles unweighted maximum parsimony (*1*), which falls prey to long-branch attraction under rate heterogeneity (*14*). Eqs. (1)–(9) prevent this artifact in three ways. First, Eq. (4) weights every shifted mutation by $w_c = \ln \frac{L_{\text{eff}} - m_c}{m_c + 1}$: on our worked 4-taxon, 16-site quartet (88.72 nats raw, 22.18 nats root, 61.02 nats star at $m_e = 0$ with $m_c = 2$ private leaf edits costing $\mathcal{L}_{\text{branch}}(2, 16) = 9.71$ nats each; Fig. S1 and Table S5), pooling a shared change from two quiet branches ($m_{\text{short}} = 1$, $w_{\text{short}} = 2.01$ nats) saves 4.03 nats versus 0.71 nats on two six-mutation branches ($w_{\text{long}} = 0.36$ nats)—a 5.6-fold positional shield matching Felsenstein's likelihood weights (*2*). Second, with $m_c = 2$ private edits per leaf ($\mathcal{L}_{\text{branch}}(1..4, 16) = 5.56, 9.71, 12.99, 15.72$ nats), one shared mutation saves 6.55 nats on the daughter branches yet costs $\mathcal{L}_{\text{branch}}(1, 16) + \ln 3 = 6.66$ nats, contracting automatically ($G_{\text{MDL}} = -0.11$ nats, $P_{\text{local}} = 0.473$), whereas $m_e = 2$ (+12.02 vs. 10.81, $G_{\text{MDL}} = +1.21$ nats, $P_{\text{local}} = 0.771$) and $m_e = 3$ (+16.69 vs. 14.08, $G_{\text{MDL}} = +2.60$ nats, $P_{\text{local}} = 0.931$; 75.11 nats resolved total) pay for the branch header. Third, continuous-time hard-CAT `GTR+CAT8/LG08+CAT8` pruning (Eq. (9)) handles multi-hit saturation directly across nucleotide and protein alignments.

Section S2: Synthetic Validation (37,020 Histories) and Model-Selection Reconciliation

Across all simulated regimes (Figs. S2 and S3, Tables S2–S5), pricing split addresses eliminates overfitting without losing power as data accumulate. On 128-taxon star bursts (`S08`), Bushwhacker emits 0.0 false splits (versus 125.0 for binary ML, 74.8 for `IQ-TREE -polytomy`, 11.9 for `FastTree SH90`, and 2.3 for `UFBoot95`), rises from 8.2% recall (16.3/197, 99.7% precision at $L = 100$ bp) to 99.5% recall (196.0/197, 100.0% precision at $L = 16{,}000$ bp), recovers 96.0% of Felsenstein-zone splits (0% LBA), maintains 99.2% sister-clade recall across $m = 0 \rightarrow 5$ convergent codons, cuts BioNJ error on 256-taxon pectinate trees, and improves out-of-sample held-out compression by +64.3 nats on multifurcations (RF = 0.38 vs. 16.0) and +400.9 nats under $m = 3$ convergence over binary ML (which exhibits +3.90 and +126.3 nats of in-sample likelihood excess). Across 44,337 tested splits, conditional local posteriors $P_{\text{local}}(e \mid T_{-e}, A)$ are well-calibrated at $P_{\text{local}} \geq 99\%$ (89.4% of splits, 98.34% true; overall ECE = 3.73%) while remaining optimistic on intermediate $[0.50, 0.99)$ splits (40.6%–80.3% true; Table S3) because P_{local} conditions on a single optimized topology T_{-e}. Across all 322 evaluated Stage 4 lateral shunts in `N1-N4`, conditional shunt posteriors yield 93.57% mean predicted probability vs. 90.99% empirical true accuracy (ECE = 4.93%, with only 3 shunts emitted

across 400 clonal null alignments, 0.0075 per dataset). Controlled four-mode ablations of Eq. (9) confirm that removing the $\frac{1}{2}\ln\max(1, N-3)$ multiplicity term (`-no-mult-pen`) increases short-gene false positives 3–8× (`S08b` 0.15 → 0.60, `S10b` 0.05 → 0.40, `P2` 0.04 → 0.12, `sim_bic-500` 0.17 → 0.22), whereas estimating per-alignment empirical rate dispersion (`-empirical-alpha`) tracks fixed $\Gamma(1,1)$ octiles within ΔnRF ≤ 0.011 across ToL supermatrices (ECE = 1.15% vs. 1.12%; on 100-locus `Anomaly-24` → `ASTRAL-IV`, default achieves nRF = 0.300 in 6.23 s CPU [0.60 s wall] vs. `IQ-TREE UFBoot95` 0.324 and `SH-aLRT80` 0.190 at 2,372 s CPU [37.5 s wall]; Table S11).

On the 7,200 empirical and 27,500 matched simulated alignments from published benchmarks (*13, 65*) (Table S4), 24,338 splits flip between BIC ModelFinder and GTR + I + G4, and 74.7% carry < 1.0 expected substitution; Bushwhacker contracts 96.4% (23,465) under deterministic heuristic search (baseline IQ-TREE model comparisons used default time-seeded runs with fast tree search for $N > 35$ and on `c0-c2`). This reconciles both studies (*82, 83*): average binary RF is nearly identical across models (RF_{bin} = 8.42 vs. 8.46 on `sim_bic`), yet on radiation-rich trees (> 50% short branches, $n = 625$), binary ML invents 22.62 false splits/tree (33.8% precision) and `-polytomy` invents 11.79 (50.2%), compared with 0.18 (98.8% precision) under Bushwhacker. Auditing the `phyml -o lr` reference protocol (*13*) further confirmed that identical alignments produce zero-length branches resolved identically by alphabetical BioNJ tie-breaking across models.

Section S3: Eukaryotic Radiations, Coalescent Species Trees, and Plant C_4 Networks

Across six multi-seed eukaryotic benchmarks (4,159 gene-alignments across 3,315 unique genes, 83,180 independent RAxML/IQ-TREE searches (*66*); Table S6), 48.2% of binary splits disagree across searches ("search-unstable splits"); Bushwhacker contracts 94.5% (155,821/164,901) while $\rho(A)$ quartiles track search stability (60.2% → 35.2%). Across all 19 eukaryotic datasets (45,205 loci (*12, 84*)), withholding $G_{MDL} \leq 0$ splits eliminates 61.5% (936,166/1,522,139) of apparent conflicts and resolves contested nodes via cumulative $\Delta\mathcal{L}_{MDL}$ (`BoroA6` Ctenophora 38.7 vs. 5.5 nats; `WhelA7` 248.7 vs. 84.5; `ChenA4` Lungfish 6,977.3 vs. 5,780.6; Archelosauria 16,234.8 vs. 1,596.3; Atlantogenata 2,364.8 vs. 769.6). Unweighted single-locus majorities ($\Delta\mathcal{L}_{MDL} > 0$, one vote per informative gene) yield the same directional preference on vertebrate and placental splits: `ChenA4` Lungfish+Tetrapods vs. Coelacanth vs. Lungfish+Coelacanth shifts from a forced-binary tie (676 : 665 : 651, $\chi^2 = 0.47, p = 0.79$) to 407 : 344 : 244 ($\chi^2 = 40.74, p = 1.4 \times 10^{-9}$), `ChenA4` Archelosauria leads 955 : 157 : 104 ($p = 2.9 \times 10^{-244}$), and `SongD1` Atlantogenata leads 98 : 35 : 21 ($p = 5.8 \times 10^{-15}$); on deep metazoan roots (`BoroA6` 4 : 1 : 0, $p = 0.074$; `WhelA7` 9 : 4 : 1, $p = 0.030$), however, MDL-resolved counts are small and Bushwhacker evaluates site-homogeneous `LG08+CAT8` models rather than site-heterogeneous empirical profile mixtures (`CAT-GTR`/PMSF) designed for deep compositional heterogeneity. Directly benchmarking bootstrap and near-zero polytomy baselines on mammalian *CHST13* (22 species, 1,176 bp; Fig. S7A–B) clarifies why support thresholding does not substitute for information-theoretic contraction: raw ML (2.94 s single-core) injects 8 conflicting splits (11 true clades), and `IQ-TREE --polytomy` (2.95 s) removes only one near-zero branch (`Gorilla+Chimp`, bl = 10^{-6}, UFB = 75%). At 1,000 `UFBoot` replicates (3.84 s), half of those false splits retain ≥ 50% support (4 FP remain, including `Mouse+Rat+Bushbaby` at 63% and `Sloth+Pika` at 62%) and 1 remains at ≥ 70%, whereas raising the gate to ≥ 95% eliminates all conflicts at the cost of collapsing true `Catarrhini` (93%), `Dolphin+Cow` (91%), and `Laurasiatheria` (70%) into 8 remaining clades. Classical 100-replicate nonparametric bootstrap requires 277.1 s single-core CPU (1 FP at ≥ 50%

and ≥ 70%, dropping to 7 true clades at ≥ 95%); SH-aLRT ≥ 80% retains 1 conflict (9 TP); aBayes ≥ 95% and FastTree SH ≥ 90% (0.20 s) match the 9 TP / 0 FP frontier; and Bushwhacker reaches the same 9 true clades (P_{local} = 0.82–1.00) with 0 conflicts in 0.020 s under --empirical-alpha (10 TP and 2 borderline splits at P_{local} = 0.664–0.690 under default GTR+CAT8, 0 FP for $P_{\text{local}} \geq 0.74$, 0.019 s).

Passing multifurcating gene trees directly into summary coalescent solvers (ASTRAL-III/ASTRAL-IV (*29, 85*)) improves coalescent species-tree inference without bootstrap cutoffs in Anomaly-Zone simulations (nRF = 0.2048 vs. 0.2429–0.2762 for FastTree across 2,000 loci; Table S2; Fig. S7C compares cutoff sensitivity across UFBoot 10–95% [0.47–0.81 s vs. 35–54 s]) and across empirical radiations spanning jawed vertebrates (ChenA4: nRF = 0.0727 vs. 0.1091 binary), Neoaves birds (PrumD6: 0.1624 vs. 0.1726), fungi (NagyA1: 0.1754 vs. 0.1930), whole-genome birds (Shultz (*38*): 0.2778 vs. 0.3056), and placental mammals (SongD1: nRF = 0.0294 matching IQ-TREE-10), while exposing the boundary on ultra-short (L = 300 bp) or Precambrian single-gene markers where contracting 70–75% of splits removes weak single-substitution signal. In land plants (WickD3a → WickD3b, 844 loci; Fig. S5D), dropping third codon positions eliminates 88.3% (60 → 7) of GC/AT SPARSE artifacts while preserving *Sorghum* ↔ *Zea* conversion (+8.04 nats); in Poaceae, Bushwhacker recovers native C_3 *Alloteropsis* paralogs inside Paniceae (P_{local} = 1.000), places horizontally transferred C_4 pck-1P1/ppc-1P3 inside donor clades (P_{local} = 1.000), and isolates post-transfer catalytic convergence (+19.66 nats across 27 ppc codons; +16.01 nats cumulative across two pck SPARSE links [+12.58 and +3.43 nats, 12 and 11 sites]).

Section S4: Mammalian *Prestin* and Proteome-Wide TOGA2 Census (19,953 Loci)

On the 116-mammal *Prestin* (*SLC26A5*, 2,235 bp) coding alignment (*19, 67*) (Fig. S4), Bushwhacker resolves 81/113 binary splits (19 polytomy hubs), maintains monophyletic Cetacea within Artiodactyla (P_{local} = 1.00) and monophyletic Chiroptera (P_{local} = 1.00; 76 TP, 5 FP, 93.8% precision, nRF = 0.2165 in 0.048 s vs. 18 FP/83.6% IQ-TREE -polytomy, 3 FP/96.5% UFBoot95 [73.5% rec.], 4 FP/95.2% SH-aLRT80 [69.9% rec.], 19 FP/83.2% raw IQ-TREE 2 [78.9 s, 0 shunts], and 18 FP/84.1% FastTree 2), while isolating one primary echolocation SPARSE lateral link (+3.91 nats, P_{local} = 98.04%) uniting Delphinidae and Rhinolophoidea bats across the four classic motor-domain codons (N7T, I384T, S392A, G601A); canonical site L497M is shared by non-echolocating *Pteropus* fruit bats and rightly filtered as an ancestral Chiroptera state.

Across TOGA2 mammalian orthologs up to 882 species (19,952 coding alignments, 19,953 proteins (*23, 68*); Fig. S5A–C and Table S7), coding contraction removes 55.5% (3.87 million) of species-tree clashes (9,876,650/14,937,066 candidate splits retained; 3,102,592 conflicting = 20.8% of all candidate binary splits and 31.4% pooled across MDL-retained splits, corresponding to a length-invariant ~ 33.2–34.0% per-gene retained-split discordance floor across 19,890 stratified loci; trimmed 21.2% on 5,058 de novo WALRUS codon MSAs), while 20-state proteins eliminate 72.7% (7.87 million) of conflicts and contract 65.0% overall (5,226,678/14,937,910 candidate splits retained; 2,952,245 conflicting = 19.8% of candidate binary splits and 56.5% of retained splits). Under Kraft-exact Eq. (7), species-tree 20-state AA inference recovers 8,670 shunts across 2,872 proteins (14.4%): 2,251 contiguous INTERVAL tracts across 1,205 proteins (6.0%, median 85 aa, 68.9% intra-family / 88.3% intra-order; enriched 6.6× in globins [40.0%], 5.4× in CYP/UGT [32.8%], 4.4× in olfactory receptors [26.3%], 3.9× in KRAB-ZNF [23.8%], and 3.0× in cadherins/keratins [18.1%] vs. 4.2% background)

and 6,419 SPARSE events across 2,374 proteins (11.9%, 38.0% cross-order = 3.26× vs. INTERVAL). Across the 27-gene core cochlear hair-cell complex (*86, 87*) (OTOG absent from hg38 v1), shunts appear in 19/27 (70.4%) free-tree AA, 10/27 (37.0%) species-tree AA (USH2A +27.6 nats, ADGRV1 +20.3, TRIOBP +18.6, CDH23 +16.8, LOXHD1 +14.1, CIB2 +9.7, WHRN +5.0, STRC +4.7, PCDH15 +3.8, TMC1 +3.2), 11/27 free-tree NT, and 6/27 species-tree NT (13/27 species-tree AA ∪ NT; 0 across all four modes on negative control ACTG1). Because these 27 cochlear proteins average 1,676.0 aa (median 900.0 aa) versus 626.5 aa proteome-wide (2.68×), their total count of 18 species-tree AA SPARSE links across 9/27 genes (plus 1 ungulate INTERVAL tract in CIB2 yielding 10/27; ADGRV1, TRIOBP, and STRC carry non-echolocator hare or primate links) tracks the length-scaled expectation (23.24 expected, 0.77×); critically, however, 9/18 (50.0%) of those cochlear SPARSE links directly unite laryngeal-echolocating bat (*Yangochiroptera* ↔ *Rhinolophoidea* across PCDH15, CDH23, LOXHD1, TMC1; plus intra-bat USH2A, STRC) or toothed-whale (CDH23) clades versus 16.6% across pre-Eq. (7) candidate AA SPARSE links (14,105 unrefined proteome links across 4,770 pre-refinement genes vs. 6,419 Eq. (7)-retained links across 2,872 proteins), a 3.02× concentration (two-sided binomial $p = 1.1 \times 10^{-3}$; 10/18 = 55.6%, $p = 1.9 \times 10^{-4}$ including pteropodid link WHRN, plus two sp_nt TMC1 links in bats and narwhal–bat). Formal taxon-pair degree-preserving Poisson and 10^4-draw Monte Carlo guild audits across all 2,442 cross-order Eq. (7)-retained SPARSE AA shunts confirm proteome-wide neutrality for echolocators (59 obs. vs. 78.4 exp., 0.75×; 1.05× family-deduplicated, $p = 0.849$; 1.11× length-adjusted; bats–whales 0.50× raw, 0.75× family-deduplicated), marine mammals (1.31×, $p = 0.164$, $q = 0.196$; 2.70× across 21 families), and subterranean blind mammals (1 hit [ZDHHC21] vs. 0.68 exp., $p = 1.0$), while revealing that raw myrmecophagous excess (4 hits vs. 0.32 exp., 12.6×, $p = 3.3 \times 10^{-4}$; 35.7× family-deduplicated, $p = 6.0 \times 10^{-6}$, $q = 1.2 \times 10^{-5}$) reflects extreme protein length (TTN, ALMS1, OR10D3, ZSCAN16, mean 11,508 aa vs. 1,235 aa) and attenuates to 1.35× after length adjustment and to 1 non-multicopy hit ($p = 0.157$, $q = 0.194$). De novo WALRUS re-alignment reproduces 73.0% (1,085/1,487) of shunted loci with 1.01× echolocator neutrality ($p = 1.0$).

Section S5: Prokaryotic Domains, Clinical Plasmids, 16S Scaling, and .bwz Mosaicism

Across 242 GTDB r95 single-gene markers (*25, 72*) (Table S8), contracting saturated deep splits cuts 42.5% (61.1% in Bacteria) of discordant branches, raises bacterial genus compatibility to 92.3% (vs. 86.0% FastTree 2), and in ar122 20-state AA space recovers 11.8% (12/102, 15 shunts [6 INTERVAL, 9 SPARSE], 249.6 nats, including DPANN Aenigmatarchaeota/Micrarchaeota ↔ Nanoarchaeota exchanges (*43*)) versus 0.0% (0/20, two-sided Fisher OR = 0.0, $p = 0.213$, $q = 0.221$) across core ribosomal/RNAP subunits (*24, 88*) (clade-level NT 0/10 vs. 3/58, $p = 1.0$). In hospital *Klebsiella pneumoniae* (*42*), the 214-kb core chromosome resolves 6/6 splits ($P_{\text{local}} = 1.00$, 0.5% lateral budget; 13/13 splits across the 16-strain supermatrix, supplemented by 222 core loci across 10 genomes), whereas the matched 882-bp *blaKPC* plasmid collapses 5/6 splits (83.3%) into *blaKPC-2* and *blaKPC-3* (C814T) horizontal hubs. Scaling Greengenes 16S rRNA across eight nested tiers ($N = 256 \rightarrow 32,768$ OTUs (*44*)) reduces supported splits from 88.5% ($N = 256$) and 91.4% ($N = 1,024$) to 76.0% ($N = 32,768$; 4,585 hubs), cuts contradicting splits per taxon group by 19.4–23.8% across five ranks relative to forced-binary trees (while forced-binary FastTreeMP retains higher strict monophyly, 61.4–75.5% vs. 32.0–44.2%), runs 4.50× faster in full pipeline (251.6 s vs. 1,133.2 s) and 11.9× faster in standalone 32-core mode (94.5 s, 4.07 GB RSS vs. 1,125.3 s, 2.18 GB RSS for FastTreeMP at $N = 32,768$), and archives data at 64.43 B/OTU (2,111,286 B, 1.08× smaller

than `xz -6`). Across Table S9, `.bwz` sequence payloads outperform general dictionary compressors by 1.01×–2.06× (1.6×–18.9× `gzip -9`; full archives 63.2 B/genome [1.25× xz, 11.5× gzip] on 10,062 SARS-CoV-2) across clonal viral, mitochondrial, plastid, and 2.22-Mb *S. pneumoniae* `PMEN1` (*73, 89*) alignments until > 1–4 whole-genome breakpoints (`Sarbeco_83` 0.71× payload [0.68× archive], `HIV1_CRF02_AG` 0.65× payload [0.63× archive]) let multi-donor `lzma -9` dictionaries (*90*) overtake single-tree coding.

Section S6: Germinal-Center B Cells and Single-Cell Human Cancer Cohorts

On 50 Galton–Watson affinity-selected GCs (90 sampled cells plus germline under 5-mer SHM (*46*); Table S2), zero-leakage Bushwhacker achieves 99.83% unique-genotype split precision (0.02 vs. 42.34 FP/GC for binary ML; 0.10 s/GC; cell-level $F_1 = 0.7533$ in 4.0 s total vs. `GCtree` $F_1 = 0.876$ zero-leakage [580.9 s] and 0.881 abundance-oracle [593.5 s], 145× faster) and recovers mutant burst degree ($\rho = 0.438$, $p = 2.9 \times 10^{-181}$), whereas across 50 neutral negative-control GCs ($\beta_{\text{bind}} = 0$), although unadjusted pooled cell counts show a mild between-GC sampling difference (1.14×, $p = 1.3 \times 10^{-3}$), paired within-GC hub comparisons remain strictly null ($n = 24$ paired Wilcoxon $p = 0.705$ abundance, $p = 0.833$ affinity). Across all 147 mouse lymph-node GCs (10,429 B cells, 9,352 unique 657-bp genotypes (*26*)) and `Clone_3-8` (66 cells (*45*)), MDL eliminates binary staircase errors to 0.00 FP/GC at cell level (93.2% precision, 57.3 s vs. 1,807.1 s `IQ-TREE 2`) and 0.007 FP/GC on unique genotypes (95.1% precision, 54.3 s vs. 1,524.6 s; 1/63 vs. 63/63 splits on `Clone_3-8`). Post hoc joining shows monotonic expansion from $k = 0$ leaves (5.1% expanded ≥ 2 cells) to $k \geq 3$ hubs (48.2%; 9.5×, paired GC Wilcoxon $n = 86$, $p = 5.9 \times 10^{-6}$, $q = 1.2 \times 10^{-5}$; exact `(GC, depth)` paired Wilcoxon $n = 85$, $p = 4.1 \times 10^{-6}$, $q = 1.1 \times 10^{-5}$; GC+depth+Hamming fixed-effects OLS $\beta = +0.205$, $p = 5.3 \times 10^{-10}$, $q = 2.1 \times 10^{-9}$; mouse $n = 25$, $p = 2.0 \times 10^{-6}$, $q = 6.8 \times 10^{-6}$), a weak continuous within-GC correlation with yeast-display $\Delta \log_{10} K_D$ affinity (pooled $\rho = +0.037$, $p = 3.6 \times 10^{-4}$; GC-level mean $\rho = +0.0372$, Wilcoxon $n = 127$, $p = 1.7 \times 10^{-3}$, $q = 2.6 \times 10^{-3}$; mouse $n = 28$, $p = 2.9 \times 10^{-3}$, whereas binary hub-vs-nonhub pooling [$p = 0.203$, $q = 0.221$] and fixed depth-stratum $\Delta \log_{10} K_D$ contrasts [$p = 0.27$–0.40] are non-significant, showing that clonal burst abundance is the primary cellular signal), and 5.6× odds enrichment in CDRs among mutations recurring across ≥ 5 mice (64.0% [16/25] vs. 24.1% [61/253], OR = 5.60, $p = 7.5 \times 10^{-5}$, $q = 1.5 \times 10^{-4}$; DMS +0.305 vs. −0.083, $p = 8.1 \times 10^{-5}$).

In human tumors with cell counts and labels withheld (Fig. S6 and Table S10), ordered 3-state compression across 168 COMPASS myeloid Tapestri runs (1,001,609 cells, 6,222 genotypes (*47*); 2.62 s) recovers 6.18× larger in vivo clone sizes at $k \geq 3$ hubs (232.8 vs. 37.7 cells; per-run Wilcoxon $n = 166$, $p = 1.7 \times 10^{-18}$, $q = 1.7 \times 10^{-17}$; per-patient $n = 127$, $p = 3.5 \times 10^{-14}$; exact `(sample, depth)` paired Wilcoxon $n = 96$, $p = 5.9 \times 10^{-12}$, $q = 2.6 \times 10^{-11}$; patient-FE + depth OLS $\beta = +1.59$, $p = 1.4 \times 10^{-28}$, $q = 2.2 \times 10^{-27}$; Spearman $\rho = 0.24$, all-0 `WT` excluded) and 90.3% (2,858/3,164) stepwise WT → HET → HOM CN-LOH transitions; across the independent 123-patient Morita AML cohort (0.128 s (*27*)), direct clone cell multiplicity shows no hub correlation ($\rho = -0.060$, $p = 0.197$), and expansion signal emerges specifically under descendant-inclusive lineage totals ($\rho = +0.331$; $n = 97$, paired Wilcoxon $p = 3.0 \times 10^{-15}$; patient-FE + depth OLS $n = 462, \beta = +0.439$, $p = 3.9 \times 10^{-6}$, $q = 1.1 \times 10^{-5}$), which structurally couple internal branching nodes to their summed descendant clones; late subclonal signaling resistance drivers enrich relative to epigenetic founders (OR = 1.79, $p = 1.9 \times 10^{-3}$, $q = 2.8 \times 10^{-3}$; parallel drivers in 44 patients). On raw Tapestri genotypes from `AML-38-001` (7,016 cells ×16 variants), Bushwhacker (evaluating unweighted unique genotypes with cell counts masked) separates parallel *IDH1* ($P_{\text{local}} = 0.996$, $k = 3$)

and *IDH2* ($P_{\text{local}} = 0.997, k = 4$) hubs downstream of *NPM1* in 3.0–4.1 ms, whereas `COMPASS` MCMC (471.4 s, 3.54 GB RSS, modeling raw cell-level read counts, VAFs, doublets, and dropout) nests *IDH2* sequentially beneath *IDH1*—highlighting complementary parsimonious genotype compression versus generative cell-level inference. Across 4,950 DLP+ scWGS copy-number genomes (4,375 500-kb bins, 12 states, evaluated against published `Sitka` phylogenies without re-executing the JVM/Nextflow pipeline (*48*)), ordered step coding finishes in 21.9 s on colinear segments (113.6 s raw bins; 21.9–33.5× raw-bin compression), contracts 33.1–49.9% (39.0–53.1% segments) of noisy binary splits while preserving cohort clade purity (57.9%–95.7% seg / 60.7%–95.7% bin vs. 20.8%–59.9% random null and 60.1%–96.7% forced-binary), isolates primary ovary vs. ascites subclones (`OV2295` 95.7% clade purity, weighted $F_1 = 0.997$ seg / 0.995 bin), and notes sample-level HGSOC aneuploidy rank in `OV2295` ($\rho = 0.818, q = 1.5 \times 10^{-266}$, an input property; TNBC PDX `SA609/SA501/SA535` $\rho = -0.11, -0.05, +0.03$, $p = 0.475$ on `SA535`).

Supplementary Figures

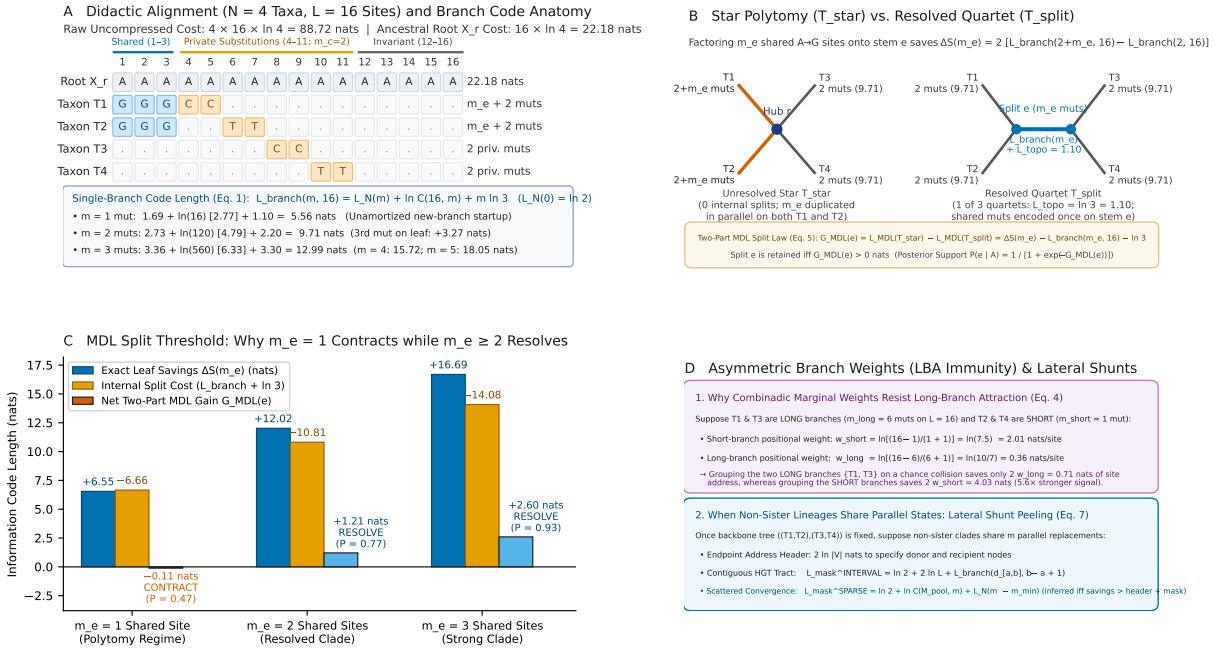


Figure S1: **Worked two-part MDL ledger on a 4-taxon, 16-site alignment** ($N = 4, L = 16$)**.** (**A**) Alignment matrix ($X_r = \mathbf{A}^{16}$, 88.72 nats raw, 22.18 nats root) and single-branch code lengths $\mathcal{L}_{\text{branch}}(m, 16) = 5.56, 9.71, 12.99$ nats ($m = 1, 2, 3$) from Eq. (1). (**B**) Unresolved star baseline T_{star} (61.02 nats) versus resolved quartet ($\mathcal{L}_{\text{topo}}(2, 2) = \ln 3 = 1.10$ nats; 75.11 nats total at $m_e = 3$). (**C**) Exact leaf savings $\Delta\mathcal{L}_{\text{children}}(m_e)$, split cost, and net gain $G_{\text{MDL}}(e)$ for $m_e = 1$ (−0.11 nats, $P_{\text{local}} = 0.47$, contracted), $m_e = 2$ (+1.21 nats, $P_{\text{local}} = 0.77$), and $m_e = 3$ (+2.60 nats, $P_{\text{local}} = 0.93$). (**D**) Combinadic positional asymmetry ($w_{\text{short}} = 2.01$ vs. $w_{\text{long}} = 0.36$ nats) and Eq. (7) shunt pricing.

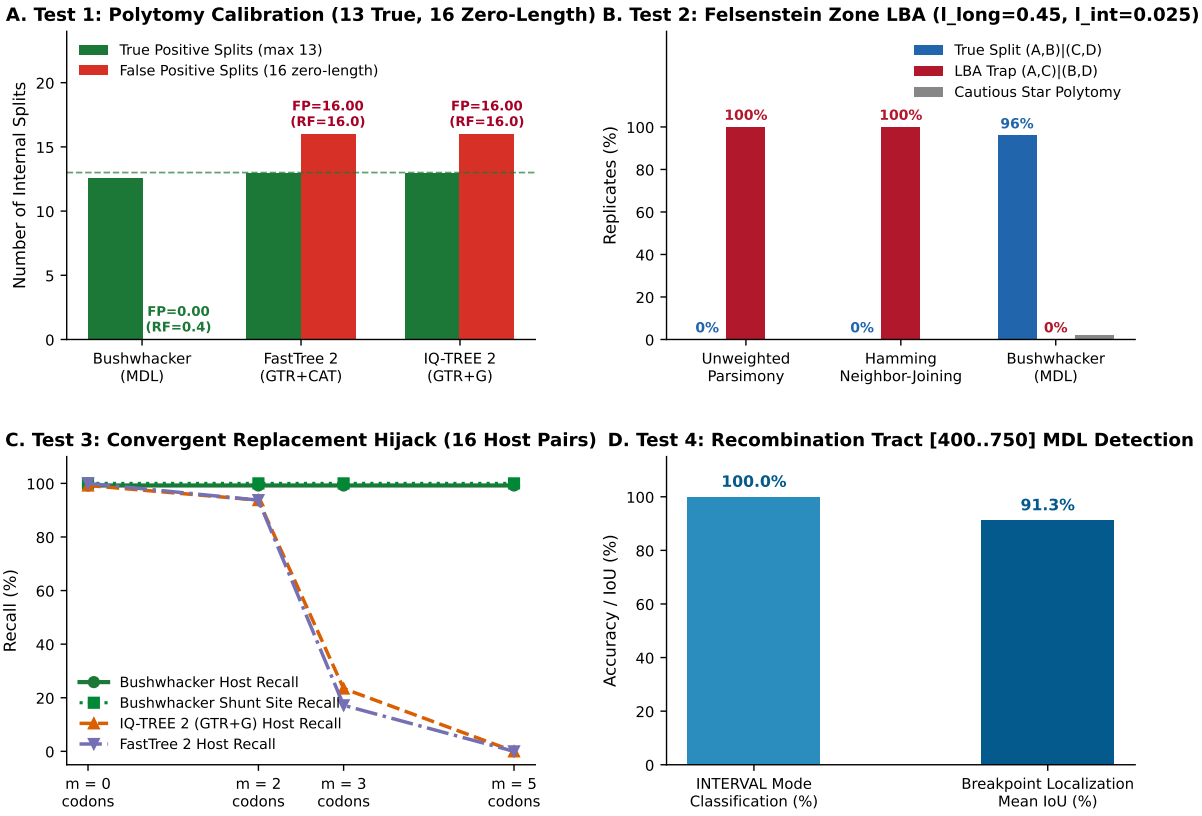


Figure S2: **Ground-truth control benchmarks across polytomy calibration, Felsenstein long-branch attraction, convergent hijack, and recombination tracts.** (**A**) Polytomy test (13 true and 16 zero-length internal splits): Bushwhacker emits 0.00 false splits (RF = 0.4) versus 16.00 false splits under forced-binary `FastTree 2` and `IQ-TREE 2`. (**B**) Felsenstein Zone ($l_{\text{long}} = 0.45$, $l_{\text{int}} = 0.025$): 96% true split recovery and 0% long-branch attraction versus 100% attraction under unweighted parsimony and neighbor-joining. (**C**) Sister-clade preservation across $m = 0 \rightarrow 5$ convergent codons (16 host pairs; 99.2% Bushwhacker recall versus 0% binary ML recall at $m = 5$). (**D**) Exact `INTERVAL` classification (100.0%) and breakpoint localization (IoU = 91.3%) on $[400, 750]$ mosaic tracts.

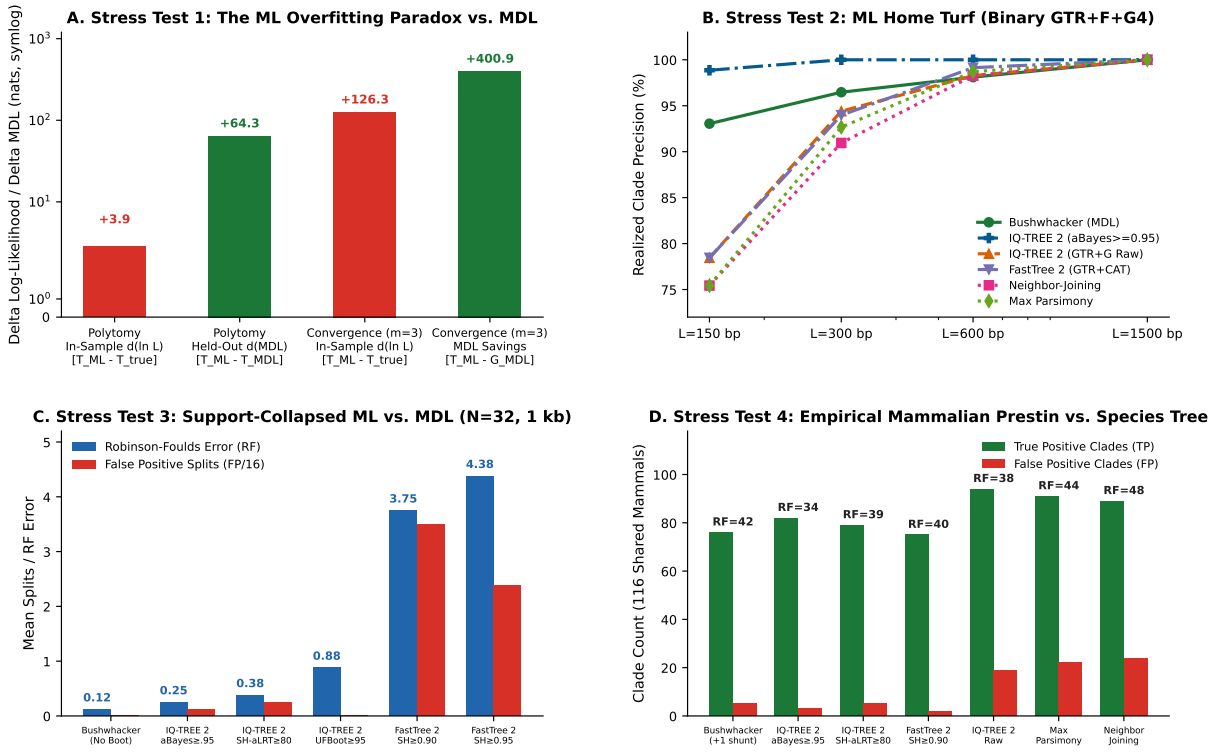


Figure S3: **Overfitting controls, support-collapsed baselines, and empirical mammalian *Prestin* comparison.** (**A**) In-sample binary ML likelihood excess (+3.90 and +126.3 nats) versus out-of-sample held-out compression advantage (+64.3 nats on multifurcations; +400.9 nats under $m = 3$ convergence). (**B**) Realized clade precision on strictly binary `GTR+F+G4` trees across $L = 150 \rightarrow 1,500$ bp. (**C**) Mean RF error and false splits for Bushwhacker (0.12 RF) versus `IQ-TREE 2` (`aBayes`, `SH-aLRT`, `UFBoot`) and `FastTree SH` cutoffs. (**D**) Empirical 116-mammal *Prestin* split recovery relative to the NCBI mammalian species tree (all 116 taxa; 76 TP, 5 FP, 93.8% precision under Bushwhacker with +1 echolocation shunt).

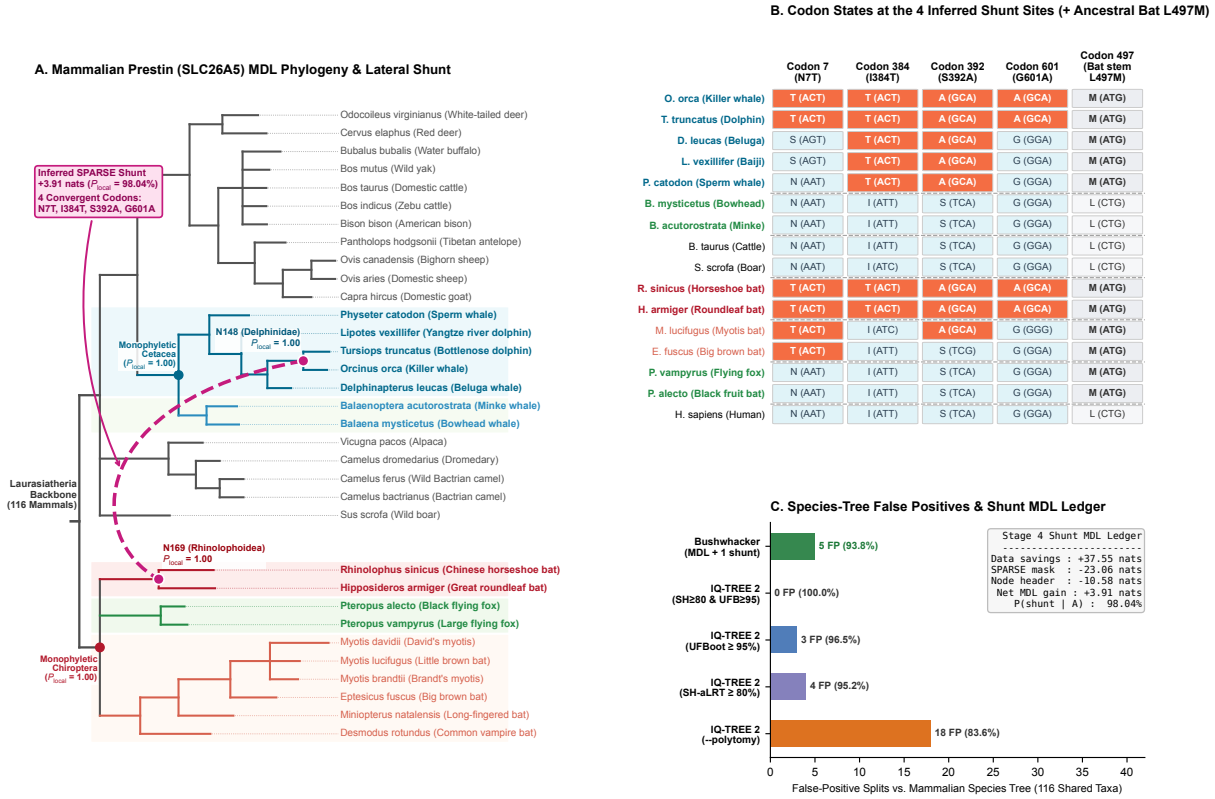


Figure S4: **Unsupervised graph induction on 116-mammal *Prestin* (SLC26A5) preserves cetacean and bat monophyly while isolating convergent echolocation codons.** (**A**) Laurasiatheria subtree showing monophyletic Cetacea within Artiodactyla ($P_{\text{local}} = 1.00$), monophyletic Chiroptera ($P_{\text{local}} = 1.00$), and the `SPARSE` lateral link (+3.91 nats, $P_{\text{local}} = 98.04\%$) uniting Delphinidae and Rhinolophoidea bats across codons `7, 384, 392, 601`. (**B**) Codon and amino-acid states across the four classic echolocation sites (`N7T`, `I384T`, `S392A`, `G601A`), contrasted with Chiroptera stem substitution `L497M` shared by non-echolocating *Pteropus* fruit bats. (**C**) False-positive species-tree splits across inference methods (5 FP, 93.8% precision for Bushwhacker vs. 3 FP/96.5% `IQ-TREE UFBoot95`, 4 FP/95.2% `SH-aLRT80`, 18 FP/83.6% `-polytomy`, and 19 FP/83.2% raw `IQ-TREE 2` [78.9 s]) and exact two-part MDL shunt ledger.

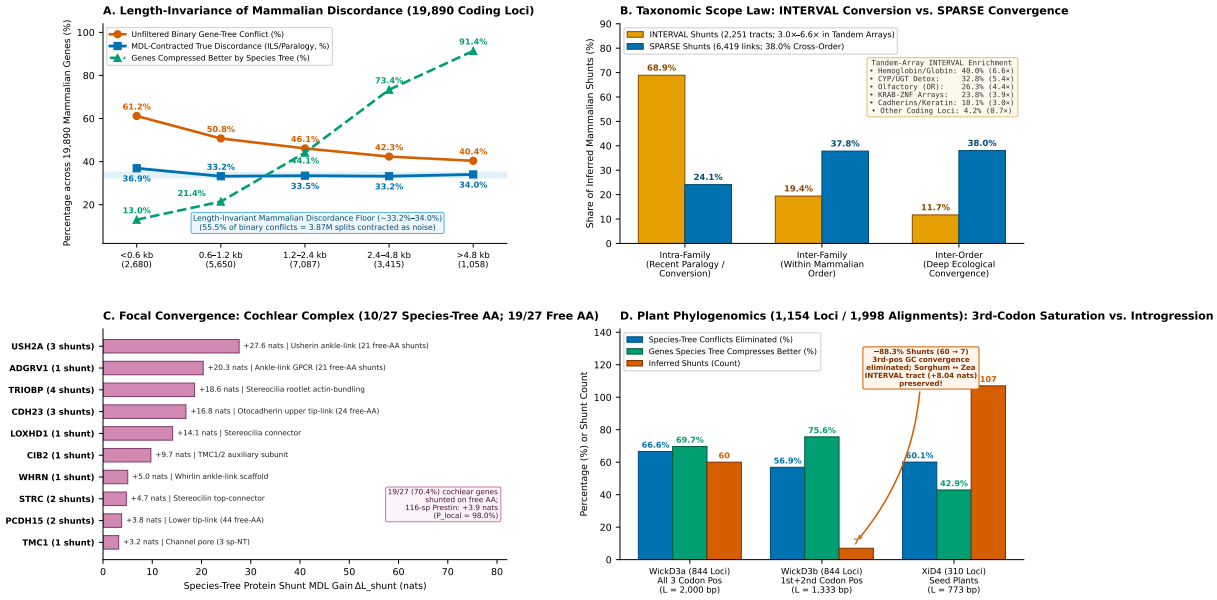


Figure S5: **Mammalian coding-DNA discordance floor, paralog conversion versus cross-order convergence, cochlear co-convergence, and land-plant saturation.** (**A**) Length stratification across 19, 890 TOGA2 coding genes showing 55.5% (3.87 million) conflict removal down to a $\sim$ 33.2–34.0% retained-split discordance floor (31.4% pooled over retained splits vs. 20.8% of all 14.94M candidate binary splits). (**B**) Taxonomic scope across 8, 670 Eq. (7)-retained species-tree 20-state AA shunts (2, 872 genes, refined from 4, 770 pre-refinement genes) separating contiguous block conversion (2, 251 tracts; 68.9% intra-family, 3.0–6.6× enriched in tandem/multicopy families vs. 4.2% background) from scattered parallel links (6, 419 events; 38.0% cross-order). (**C**) Species-tree 20-state AA shunt gains across the 10/27 shunted core cochlear hair-cell proteins (`USH2A` +27.6 nats, `ADGRV1` +20.3 nats, `TRIOBP` +18.6 nats, `CDH23` +16.8 nats, `LOXHD1` +14.1 nats; 19/27 in free-tree AA; 116-mammal coding *Prestin* +3.91 nats); although total cochlear counts follow protein length (mean 1, 676 vs. 626.5 aa; 18 `SPARSE` vs. 23.2 expected), 9/18 (50.0%) of species-tree `SPARSE` links directly unite echolocating bat or toothed-whale clades (3.02×, $p = 1.1\times10^{-3}$). (**D**) Land-plant all-codon (`WickD3a`) versus codons 1 + 2 (`WickD3b`) eliminating 88.3% (60 $\rightarrow$ 7) of GC/AT artifacts while keeping *Sorghum–Zea* conversion (+8.04 nats).

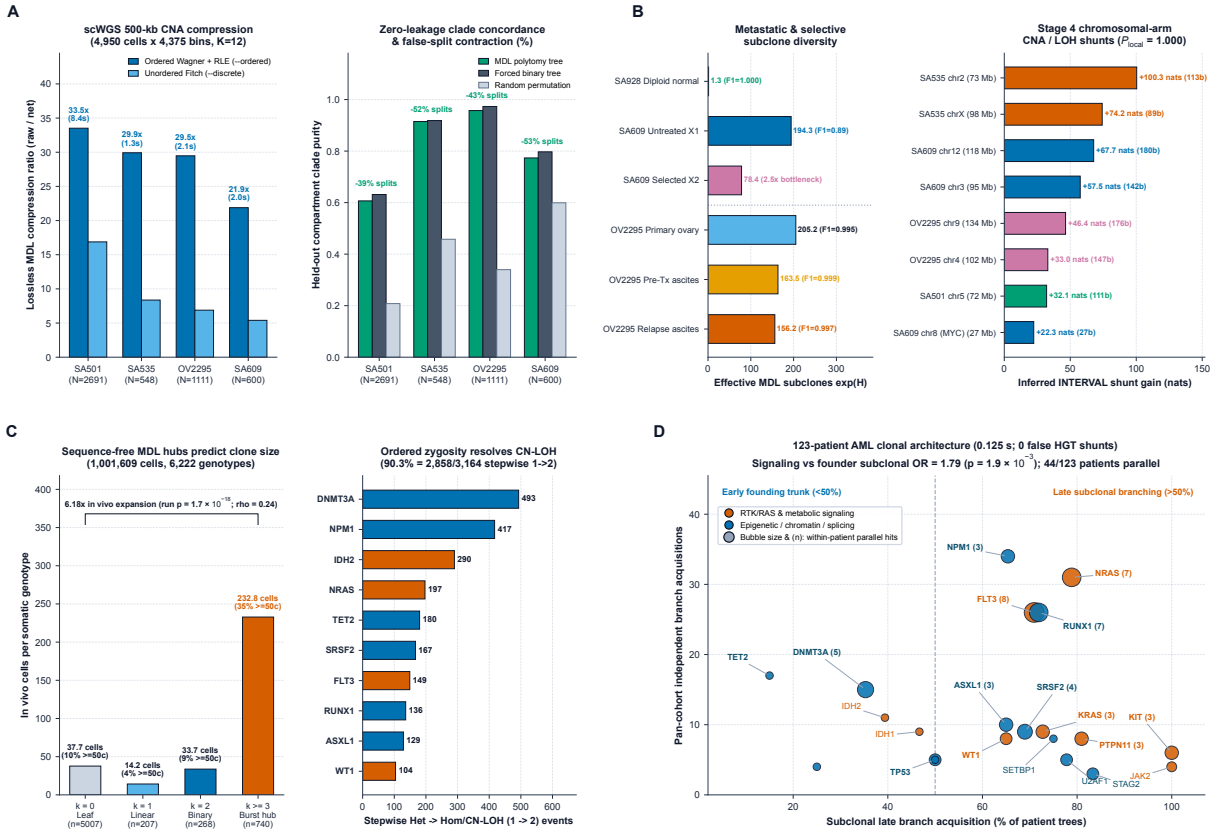


Figure S6: **Zero-leakage single-cell cancer phylogenomics across 4,950 DLP+ whole-genome copy-number profiles and 1,001,609 myeloid cells.** (**A**) Ordered 12-state copy-number compression (21.9–33.5× raw bins across `SA501`, `SA535`, `OV2295`, `SA609`) alongside held-out anatomic/passage clade purity (`OV2295` 95.7% purity, $F_1 = 0.997$ seg / 0.995 bin) and 33.1–49.9% (39.0–53.1% segments) false-split contraction. (**B**) Subclonal diversity bottlenecks and inferred chromosomal-arm CNA/LOH tracts including `MYC` and `TP53`. (**C**) Across 168 Tapestri myeloid runs (1, 001, 609 cells, 6, 222 clones (*47*)), inferred hub out-degree $k \geq 3$ predicts 6.18× larger in vivo clone sizes (per-run Wilcoxon $p = 1.7 \times 10^{-18}, q = 1.7 \times 10^{-17}$; $\rho = 0.24$) and captures 90.3% (2, 858/3, 164) stepwise WT $\rightarrow$ HET $\rightarrow$ HOM CN-LOH transitions. (**D**) Trunk-versus-subclone driver landscape across 123 AML patients (*27*) (OR = 1.79 late signaling enrichment, $p = 1.9 \times 10^{-3}, q = 2.8 \times 10^{-3}$; parallel acquisitions in 44 patients).

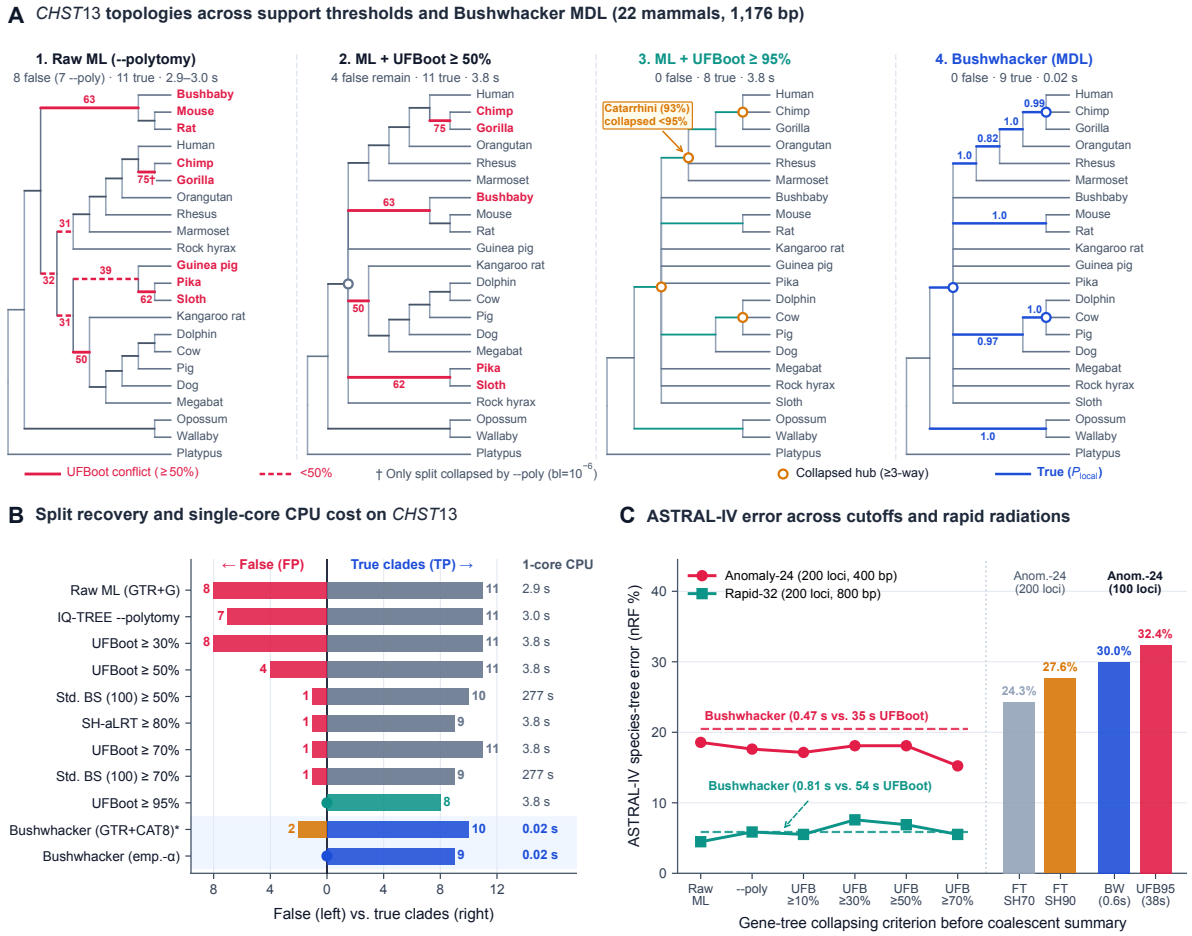


Figure S7: **Support-thresholded maximum likelihood versus deterministic MDL contraction on mammalian *CHST13* and multi-locus `ASTRAL-IV` radiations.** (**A**) Side-by-side cladograms on the 22-mammal *CHST13* alignment (1, 176 bp; main-text Fig. 3A) comparing raw `IQ-TREE` (`GTR+G`, 8 false splits in crimson; `--polytomy` removes only `Gorilla+Chimp`, bl = 10^{-6}, marked †), `UFBoot` ≥ 50% (4 conflicts remain at 50–75%), `UFBoot` ≥ 95% (0 conflicts, but collapses true `Catarrhini` [93%] into an amber polytomy alongside `Dolphin+Cow` [91%]), and Bushwhacker (`--empirical-alpha`, 0 conflicts, 9 true clades at P_{local} = 0.82–1.00 in 0.02 s). (**B**) Precision–recall–runtime frontier across ML collapsing rules (1-core CPU, including 100-replicate nonparametric bootstrap at 277 s); asterisk marks default `GTR+CAT8` (10 TP, plus 2 borderline splits at P_{local} = 0.664–0.690 and 0 FP for P_{local} ≥ 0.74, 0.02 s). (**C**) Multi-locus `ASTRAL-IV` species-tree error (nRF %) across `UFBoot` collapse cutoffs on 200-locus `Anomaly-24` (400 bp) and `Rapid-32` (800 bp) simulations (0.47–0.81 s Bushwhacker vs. 35–54 s `UFBoot` on 64 cores), contrasted on the right with `FastTree SH90` (27.6%) and 100-locus `Anomaly-24 UFBoot` ≥ 95% (32.4% vs. 30.0% Bushwhacker in 0.60 s; collapsing IQ-TREE 2 by `SH-aLRT` ≥ 80% reaches 19.0% nRF at 37.5 s).

Supplementary Tables

Table S1: **Master overview of empirical biological corpora and ground-truth simulation suites evaluated with Bushwhacker.** All single-cell abundance counts, DMS binding affinities, clinical timepoints, and taxonomic labels were strictly withheld during tree and network inference.

Biological Scale / Benchmark Suite	Alphabet / Mode	Datasets / Runs	Taxa or Cells ($\times L$)	Primary Biological / Statistical Target	Key Finding
Model-Selection Archive (*13, 65*)	Nucleotide (4-state)	7, 200	7–150 × 100–8,000 bp	BIC vs. GTR+I+G4 vs. JC discordance	96.4% of model-conflicting splits contracted
Eukaryotic Radiations (*12, 38, 66*)	NT & 20-state AA	45, 255 (45, 205 + 50 avian; 11, 164 `ASTER`)	20 corpora (19 + 1; 83, 180 ML runs)	Gene-tree conflict & `ASTRAL-IV`	94.5% search-unstable & 61.5% sp. conflict cut
Mammalian Proteome (TOGA2 (*23*))	Codon NT & 20-state AA	39, 905 (19, 952 NT + 19, 953 AA)	Up to 882 sp. (20 orders)	Species concordance, conversion, convergence	72.7% AA conflict cut; 3.0–6.6× conversion; 1.35× len-adj. guilds
Mammalian Supermatrix (*23*)	GTR+CAT8 & LG08+CAT8	2 (253 loci)	882 × 543,864 bp/181,288 AA	Deep mammalian Tree of Life	92.7% IQ-TREE agreement; 24–37× faster (193–288× FT)
Prokaryotic Domains (GTDB r95 (*25*))	NT, AA, & LG08+CAT8	434 + 2 supermatrices	1, 672 Archaea; 1, 000 Bacteria	Precambrian phyla & Complexity Hypothesis	82.4–94.3% phyla; 11.8% vs. 0% ar122 HGT ($p = 0.213$)
Botanical & Poaceae C_4 (*32, 40*)	Codon NT & 20-state AA	2, 002	Up to 483 plant species	3rd-codon GC/AT bias & C_4 HGT + conv.	88.3% 3rd-pos. shunts removed; C_4 HGT+conv.
Clinical *K. pneumoniae* (*42*)	NT & 20-state AA	9–16 str.; 222 loci	9–16 genomes (214 kb + *blaKPC*)	Chromosomal core vs. resistance plasmid	99.5% vertical core vs. 83.3% star plasmid
Pandemic Viruses (SARS-CoV-2, DENV, HIV)	NT, AA, & `.bwz`	16 cohorts	10, 062 SC2; 5, 498 DENV-1..4	Superspreading hubs, recombinants, `.bwz`	38.5 B payload / 63.2 B arch. (18.9×/11.5× gzip); 76,000× GARD
Microbiome 16S rRNA (*44*)	Nucleotide & `.bwz`	8 scaling tiers	256 → 32, 768 × 1,348 bp	Single-marker topological saturation	91.4% → 76.0% supported; 4.5–11.9× FastTreeMP
Germinal-Center B Cells (*26, 45*)	Nucleotide (deduplicated)	148 GCs (147 + 1)	10, 495 B cells (9, 394 geno.)	In vivo clonal bursts & DMS K_D affinity	9.5× expansion ($p = 5.9 \times 10^{-6}$); FE OLS $p = 5.3 \times 10^{-10}$
Human Myeloid scDNA (*27, 47*)	Ordered 3-state zygosity	291 (168 runs + 123 pts)	1, 001, 609 cells (6, 222 clones)	Leukemic expansion, CN-LOH, late signaling	6.18× hub cells ($p = 1.7 \times 10^{-18}$); FE $p = 1.4 \times 10^{-28}$; 90.3% LOH
Human Tumor DLP+ scWGS (*48*)	Ordered 12-state CNA	4 cohorts (seg + bin)	4, 950 cells ×4,375 500-kb bins	Subclonal aneuploidy & ascites evolution	21.9 s seg (21.9–33.5× comp.); $F_1 = 0.997$ purity
Ground-Truth Simulations (*13, 80*)	NT, AA, MSC, Net, Somatic	36, 920 (+ 100 GCs)	7–1,024 taxa; $L = 40$–16,000	Polytomies, calibration, HGT, single-cell	99.7–100% burst prec.; 3.73% ECE; 0% HGT FPR

Table S2: **Multi-alphabet ground-truth validation across 20-state proteins, Anomaly-Zone multi-species coalescence, reticulate networks, and single-cell somatic profiles** (9, 300 **histories; complementing the** 27, 720 **nucleotide alignments in Tables S4 and S5).**

Suite & Ground-Truth Regime	Method / Configuration	Res / True	FP / Tree	Precision	Recall	nRF / Auxiliary Metric	Time (s)
Protein Suite (LG+G4, $N = 64$, $L = 400$ AA)							
P1 Star Burst (100 reps, 0 true splits)	Bushwhacker (`-protein` MDL)	0.05/0.0	0.05	96.0% (0-FP)	—	0.08% FPR	0.81
	IQ-TREE 2 (`LG+G4 -polytomy`)	38.67/0.0	38.67	0.0%	—	63.4% FPR	9.08
	IQ-TREE 2 / FastTree 2 (Binary)	61.00/0.0	61.00	0.0%	—	100.0% FPR	9.11/4.59
P2 Rapid Radiation (50% silent stems)	Bushwhacker (`-protein` MDL)	30.14/30.49	0.05	99.8%	98.6%	nRF = 0.0077	1.39
	IQ-TREE 2 (`-polytomy`)	50.31/30.49	19.84	60.5%	99.9%	nRF = 0.3362	8.06
	IQ-TREE 2 / FastTree 2 (Binary)	61.00/30.49	30.55/30.54	49.9%	99.9%	nRF = 0.5189	8.01/3.84
P3 Standard Resolved Yule (61 splits)	Bushwhacker (`-protein` MDL)	59.60/61.0	0.01	100.0%	97.7%	nRF = 0.0116	0.99
	IQ-TREE 2 (`LG+G4`)	61.00/61.0	0.08	99.9%	99.9%	nRF = 0.0013	7.04
P4 Convergent Protein Sites (45 splits)	Bushwhacker (`-protein` + Stage 4)	44.40/45.0	0.01	100.0%	98.6%	92.8% conv. AA site prec.; 0% false clade	0.89
Coalescent MSC & B-Cell GC Suite (8, 100 trees)							
24-sp. Asymmetric Anomaly Zone ($L = 400$ bp)	Bushwhacker (MDL) → ASTRAL-IV	37.8% gene res.	—	—	—	Species nRF = 0.2048 (−15.6% vs. FT)	0.47
	FastTree 2 (raw / SH-collapsed) → ASTRAL	100% / SH	—	—	—	Species nRF = 0.2429–0.2762	2.10
32-sp. Rapid Radiation ($L = 800$ bp)	Bushwhacker (MDL) → ASTRAL-IV	54.1% gene res.	—	—	—	Species nRF = 0.0586 (ties IQ `-poly`)	0.62
50-Galton–Watson B-Cell GCs (90 cells/GC)	Bushwhacker (MDL)	7.34/35.26	0.02	99.8%	20.7%	$F_1 = 0.3356$ (0.7533 cell); burst $\rho = 0.438$	0.10
	`GCtree` (zero-leakage / abund.-oracle)	38.10/37.86	5.96/5.62	84.4%/85.2%	91.2%/91.4%	Unique geno. $F_1 = 0.8760/0.8813$	11.62/11.87
50 Neutral B-Cell GCs ($\beta_{\text{bind}} = 0$ control)	Bushwhacker (MDL)	8.58/33.92	0.04	99.5%	25.1%	Null paired hub abund. $p = 0.705$, affin. $p = 0.833$	0.05
Reticulate Network Suite (400 sims, 45 splits)							
`N1` Strictly Clonal Null (100 reps; 400 all nulls)	Bushwhacker (MDL + Stage 4)	44.98/45.0	0.00	100.0%	99.96%	0.0% on `N1` (3/400 = 0.75% across all nulls)	0.09
`N2` Short Contiguous HGT (200 bp)	Bushwhacker (MDL + Stage 4)	44.96/45.0	0.09	99.8%	99.7%	37.0% `INTERVAL` (IoU = 0.274; 0% false clade)	0.11
`N3` Contiguous HGT Tract (450 bp)	Bushwhacker (MDL + Stage 4)	44.94/45.0	1.34	97.0%	96.9%	90.0% `INTERVAL` (IoU = 0.715; 0% false clade)	0.12
`N4` Scattered Convergence (10 NT sites)	Bushwhacker (MDL + Stage 4)	44.95/45.0	0.00	100.0%	99.9%	92.0% `SPARSE`; 99.2% site prec.; 90.9% rec.	0.10
Single-Cell Somatic Suite (400 matrices)							
`C1` Binary SNV Clonal Burst (5% dropout)	Bushwhacker (`-discrete` MDL)	33.34/33.47	4.56	86.1%	85.6%	6.3× lower FP than binary ML (nRF = 0.1418)	0.035
	IQ-TREE 2 (Binary `GTR2/BIN`)	61.00/33.47	28.72	52.9%	96.4%	nRF = 0.4614	0.45
	`COMPASS` (MCMC)	12.90/33.47	5.53	57.1%	22.1%	nRF = 0.4733	30.93
`C2` High Allelic Dropout (20% missing)	Bushwhacker (`-discrete` MDL)	36.41/37.15	13.33	63.2%	61.9%	nRF = 0.3725 (vs. 0.4830 IQ-TREE BIN)	0.065
	`COMPASS` (MCMC)	13.70/37.15	6.87	49.7%	18.4%	nRF = 0.5011	27.91
`C3` Ordered Copy-Number ($K = 7$ states 0..6)	Bushwhacker (`-ordered` L_1 step)	59.60/55.60	5.61	90.6%	97.1%	nRF = 0.0660 (46% lower nRF vs. binarized)	0.14
	Binarized Copy-Number Baseline	52.37/55.60	5.15	90.1%	84.9%	nRF = 0.1224 (RF = 13.53 vs. 7.22)	0.14
`C4` Noisy Diploid Null (100 controls)	Bushwhacker (`-ordered` MDL)	0.00/0.00	0.00	100.0%	100.0%	0.0 false clones across all 100 controls	0.11

Table S3: **Conditional local posterior calibration across** 44, 337 **inferred branches and** 322 **lateral shunts without bootstrap resampling.** Conditional local posterior probability $P_{\text{local}} = (1+e^{-G_{\text{MDL}}})^{-1}$ is compared directly against ground-truth bipartition membership (Panel A; mean predicted P_{local} = 98.08%, empirical true fraction = 94.35%, ECE = 3.73%) and Stage 4 lateral shunt membership across `N1–N4` (Panel B; 322 shunts, mean predicted 93.57%, true = 90.99%, ECE = 4.93%, with 3 null shunts across 400 clonal alignments = 0.0075/dataset).

Conditional Local Posterior Bin $P_{\text{local}}(\cdot \mid T_{-}, A)$	Features (n)	Mean Predicted P_{local}	Empirical True Fraction	Absolute Gap
Panel A: Internal Tree Branches (Multi-Alphabet)				
[0.50, 0.65)	881	57.60%	40.64%	16.96%
[0.65, 0.80)	959	72.87%	48.80%	24.07%
[0.80, 0.90)	807	85.10%	57.25%	27.85%
[0.90, 0.95)	718	92.65%	69.36%	23.29%
[0.95, 0.99)	1, 353	97.35%	80.27%	17.08%
[0.99, 1.00]	39, 619	99.98%	98.34%	1.64%
Panel A Total / Weighted Summary	**44, 337**	**98.08%**	**94.35%**	**ECE = 3.73%**
Panel B: Stage 4 Lateral Shunts (`N1–N4`)				
[0.50, 0.65)	20	57.55%	75.00%	17.45%
[0.65, 0.80)	20	73.32%	70.00%	3.32%
[0.80, 0.90)	22	84.26%	77.27%	6.99%
[0.90, 0.95)	27	92.88%	88.89%	3.99%
[0.95, 0.99)	47	97.23%	97.87%	0.64%
[0.99, 1.00]	186	99.90%	95.16%	4.74%
Panel B Total / Weighted Summary (0.0075 **null shunts/tree)**	**322**	**93.57%**	**90.99%**	**ECE = 4.93%**

Table S4: **Ground-truth accuracy across** 27, 500 **empirical-parameterized nucleotide histories from published simulation archives (*13, 65*).** Panel A stratifies `sim_bic` (n = 5, 901) by short ancestral branches ($\ell_e L < 1.0$ expected substitution). Panel B reports homogeneous (`c0`), site-heterogeneous (`c1`), and branch-heterogeneous (`c2`) corpora.

Benchmark Stratum	Method	Res / True	RF_{bin}	nRF_{bin}	$RF_{\geq 1}$	$nRF_{\geq 1}$	FP / Tree	Precision
Panel A: `sim_bic` All (n = 5, 901)	Bushwhacker (MDL)	8.9/18.1	10.51	0.2727	6.87	0.2303	0.68	94.7%
	IQ-TREE 2 (BIC ModelFinder)	18.1/18.1	8.42	0.1666	5.75	0.1616	4.21	83.3%
	IQ-TREE 2 (GTR + I + G4)	18.1/18.1	8.46	0.1678	5.77	0.1623	4.23	83.2%
	IQ-TREE 2 (Jukes–Cantor)	18.1/18.1	8.99	0.1782	6.33	0.1739	4.49	82.2%
	IQ-TREE 2 (`-polytomy`)	15.4/18.1	6.82	0.1404	3.34	0.1147	2.08	89.2%
	FastTree 2	16.7/18.1	7.24	0.1542	4.56	0.1516	2.93	84.9%
	BioNJ	16.7/18.1	11.57	0.2642	8.74	0.2710	5.10	73.1%
Low Stemminess (25–50% $\ell_e L < 1$, n = 662)	Bushwhacker (MDL)	10.7/22.0	11.83	0.2744	3.72	0.1522	0.29	98.1%
	IQ-TREE 2 (`-polytomy`)	14.2/22.0	12.05	0.2679	4.38	0.1650	2.14	86.9%
	FastTree 2	20.5/22.0	15.20	0.3306	10.23	0.3150	6.88	68.1%
	IQ-TREE 2 (BIC / GTR + I + G4)	22.0/22.0	16.51	0.3535	11.57	0.3388	8.26	64.7%
Very Low Stemminess (> 50% $\ell_e L < 1$, n = 625)	Bushwhacker (MDL)	7.8/33.6	26.14	0.3961	3.69	0.1797	0.18	98.8%
	FastTree 2	22.3/33.6	33.95	0.4992	15.12	0.4964	11.34	51.4%
	IQ-TREE 2 (`-polytomy`)	20.9/33.6	36.27	0.5388	14.20	0.4540	11.79	50.2%
	IQ-TREE 2 (BIC / GTR + I + G4)	33.6/33.6	45.23	0.6623	26.47	0.6352	22.62	33.8%
Panel B: `c0` Homogeneous (n = 7, 199)	Bushwhacker (MDL)	8.7/16.8	13.30	0.3328	9.21	0.3074	2.60	82.5%
	IQ-TREE 2 (`-polytomy` / GTR + I + G4)	14.3/16.8	14.70/16.43	0.3015/0.3337	10.79/12.97	0.2865/0.3235	6.14/8.21	71.6%/66.6%
`c1` Site-Heterogeneous (n = 7, 200)	Bushwhacker (MDL)	8.8/16.8	10.19	0.2578	6.11	0.2247	1.09	92.6%
	IQ-TREE 2 (`-polytomy` / GTR + I + G4)	14.2/16.8	10.30/12.04	0.1843/0.2158	6.48/8.89	0.1647/0.2133	3.85/6.02	84.9%/78.4%
`c2` Branch-Heterogeneous (n = 7, 200)	Bushwhacker (MDL)	8.8/16.8	10.25	0.2616	6.16	0.2306	1.12	92.4%
	IQ-TREE 2 (`-polytomy` / GTR + I + G4)	14.2/16.8	10.36/12.08	0.1867/0.2180	6.55/8.92	0.1682/0.2163	3.91/6.04	84.6%/78.2%

Table S5: **Worked 4-taxon, 16-site MDL ledger and representative `AliSim` operating characteristics.** Panel A gives exact nat-by-nat code lengths on a quartet ($N = 4, L = 16$, $m_c = 2$ private leaf edits costing $\mathcal{L}_{\text{branch}}(2, 16) = 9.71$ nats each; matching Fig. S1). Panel B reports representative `AliSim` regimes (10 replicates each across 22 scenarios; 220 alignments total (*64, 80*)).

Panel A: 4-Taxon Quartet Ledger ($L = 16$)	Shared Synapomorphies	Leaf Savings $\Delta\mathcal{L}_{\text{children}}$	Branch Code $\mathcal{L}_{\text{branch}}$	Topo Cost $\mathcal{L}_{\text{topo}}(2, 2)$	Net Gain $G_{\text{MDL}}(e)$	Local Post. P_{local}	Decision
$m_e = 0$ (Zero shared mutations)	`. . .` | 2 priv.	0.00 nats	−0.69 nats	−1.10 nats	−1.79 nats	0.143	Contract (T_{star})
$m_e = 1$ (Single shared site)	`G . .` | 2 priv.	+6.55 nats	−5.56 nats	−1.10 nats	−0.11 nats	0.473	Contract (T_{star})
$m_e = 2$ (Two shared sites)	`G G .` | 2 priv.	+12.02 nats	−9.71 nats	−1.10 nats	+1.21 nats	0.771	Resolve (T_1, T_2)
$m_e = 3$ (Three shared sites)	`G G G` | 2 priv.	+16.69 nats	−12.99 nats	−1.10 nats	+2.60 nats	0.931	Resolve (T_1, T_2)
Panel B: AliSim Regime (N, L)	**Method**	**Res / True**	**FP**	**Precision**	**RF**	**nRF / FPR**	**Time (s)**
`S08` Star Null (128 × 1,000 bp)	Bushwhacker (MDL)	0.0/0	0.0	100.0%	0.0	0.0% FPR	0.55
	IQ-TREE 2 (`-polytomy` / Binary)	74.8/125.0	74.8/125.0	0.0%	74.8/125.0	59.8%/100% FPR	482.5/391.6
	FastTree 2 (raw / SH≥ 0.90)	124.6/11.9	124.6/11.9	0.0%	124.6/11.9	99.7%/9.5% FPR	3.02
`S08b` Ultra-Short (200 × 100 bp)	Bushwhacker (MDL)	16.3/197	0.1	99.7%	180.9	0.4591	0.37
	IQ-TREE 2 (`-polytomy` / Binary)	142.4/197.0	31.3/74.0	78.1%/62.4%	117.2/148.0	0.2975/0.3756	96.2/91.1
`S10b` Standard (200 × 1,000 bp)	Bushwhacker (MDL / Binary)	157.2/197.0	0.0/8.6	100.0%/95.6%	39.8/17.2	0.1010/0.0437	0.63
	IQ-TREE 2 (`-polytomy` / Binary)	192.8/197.0	6.7/9.0	96.5%/95.4%	17.6/18.0	0.0447/0.0457	64.1/65.4
`S12c` Long Gene (200 × 16,000 bp)	Bushwhacker (MDL)	196.0/197	0.0	100.0%	1.0	0.0025	2.65
`S04` Large Tree (1,024 × 1,000 bp)	Bushwhacker (MDL)	784.9/1,021	0.1	99.99%	236.3	0.1157	4.22
	IQ-TREE 2 (`-polytomy` / Binary)	1,000.2/1,021.0	37.4/50.1	96.3%/95.1%	95.6/100.2	0.0468/0.0491	17.6/17.9
	BioNJ (*91*)	1,021.0/1,021	134.2	86.9%	268.4	0.1314	0.10
`S13` Low-Div Outbreak (200 × 1,000 bp)	Bushwhacker (MDL)	102.5/197	0.0	100.0%	94.5	0.2398	0.32
	IQ-TREE 2 (`-polytomy` / Binary)	155.6/197.0	3.6/33.8	97.7%/82.8%	48.6/67.6	0.1234/0.1716	62.9/62.5
	FastTree 2	196.9/197	32.9	83.3%	65.9	0.1673	3.36

Table S6: **Multi-seed maximum-likelihood optimization instability and apparent species-tree conflict removal across eukaryotic phylogenomic radiations.** Top rows report search-unstable ML splits across 20 independent RAxML/IQ-TREE optimization starts per gene on 4, 159 gene-alignments (3, 315 unique genes; `WickD3a`/`WickD3b` paired codon views) (83, 180 searches (*66*)). Bottom summary encompasses all 19 benchmarks (45, 205 loci (*12, 66*)).

Dataset ($n_{\text{genes}}, \tilde{N}, \tilde{L}$)	Mean $\rho(A)$	Search-Unstable ML Splits	Contracted by BW (MDL)	Method	Res / Sp	Discordant Splits	Concordance Prec. (nRF_{sp})
`SongD1` Mammals (424, 37, 3,157 bp)	12.53%	1, 715/14, 416 (11.9%)	1, 612/1, 715 (94.0%)	Bushwhacker / Best-40 ML	22.3/34.0	6.82/9.04	69.4%/73.4%
`XiD4` Seed Plants (310, 33, 773 bp)	8.62%	2, 715/9, 357 (29.0%)	2, 555/2, 715 (94.1%)	Bushwhacker / Best-40 ML	16.9/30.2	7.46/13.63	56.7%/55.3%
`PrumD6` Birds (259, 199, 1,524 bp)	36.92%	18, 815/50, 762 (37.1%)	16, 188/18, 815 (86.0%)	Bushwhacker / Best-40 ML	141.4/196.0	41.53/73.54	70.7%/62.4% (0.352 vs. 0.376)
`WickD3a` Plants 1+2+3 (844, 49, 2,000 bp)	9.60%	8, 378/38, 638 (21.7%)	8, 101/8, 378 (96.7%)	Bushwhacker / Best-40 ML	19.9/45.8	11.26/19.58	46.9%/57.9%
`WickD3b` Plants 1+2 (844, 49, 1,333 bp)	14.11%	11, 118/38, 638 (28.8%)	10, 504/11, 118 (94.5%)	Bushwhacker / Best-40 ML	24.4/45.8	13.81/21.55	45.5%/54.1%
`MisoD2a` Insects (1, 478, 132, 1,208 bp)	8.87%	122, 160/190, 030 (64.3%)	116, 861/122, 160 (95.7%)	Bushwhacker / Best-40 ML	75.5/128.6	58.79/81.40	23.0%/36.4%
All 6 Multi-Search Sets (4, 159 **alns,** 3, 315 **genes)**	**12.18**%	**164**, **901**/**341**, **841** (**48.2**%)	**155**, **821**/**164**, **901** (**94.5**%)	Bushwhacker / Best-40 ML	48.2/82.2	29.82/43.79	42.6%/51.2%
All 19 Eukaryotic Sets (45, 205 **loci)**	—	9 AA + 10 NT clades	61.5% sp. conflicts cut	1, 522, 139 → 585, 973	936, 166 cut	74.9% birds; 82.5% animals	20.1–20.2× AA speedup

Table S7: **Proteome-wide mammalian concordance, cochlear and ecological convergence** (19, 952 **coding /** 19, 953 **protein TOGA2 loci across** 882 **mammals), and pandemic viral recombinant mapping.** Panel A summarizes mammalian coding-DNA and 20-state protein concordance (*23, 68*). Panel B highlights mammalian adaptive convergence, tandem paralog conversion, and unsupervised viral recombinants. In Panel A, `Cand. / Ret. Conflict%` gives pooled discordant splits divided by candidate binary splits (14.94M) and by retained splits, respectively (with per-gene mean retained conflict in parentheses where computed).

Panel A: Mammalian Proteome Mode	Alignment / Tree Mode	Loci (n)	Total Splits	Sp. Shared	Sp. Conflict	Cand. / Ret. Conflict%	Conflicts Removed	Inferred Shunts (`INT` / `SPARSE`)
Coding DNA	Forced-binary tree	19, 952	14, 937, 066	7, 960, 476	6, 976, 590	46.7%/46.7%	Baseline	—
(med. $\tilde{N} = 855$, $\tilde{L} = 1{,}383$ bp)	MDL-contracted tree	19, 952	9, 876, 650	6, 774, 058	3, 102, 592	20.8%/31.4% (33.8%)	−3, 873, 998 (55.53%)	1, 329 / 3, 228 (species tree)
20-State Protein (`-translate`)	Forced-binary / MDL-contracted	19, 953	5, 226, 678	2, 274, 433	2, 952, 245	19.8%/56.5%	−7, 872, 568 (72.73%)	2, 251 / 6, 419 (species tree)
Matched Alignment Subset	De Novo Codon MSA (`MALRUS`)	5, 058	1, 537, 531	1, 135, 904	401, 627	21.7%/26.1% (28.7%)	−107, 754 (21.15%)	73.0% (1, 085/1, 487) AA shunts reprod.
Panel B: Biological Target / Cohort	**Biological System / Guild**	**$N \times L$**	**Mode**	**Shunts**	**Top Gain / Span**	**Statistical Enrichment / Runtime**	**Primary Tract or Convergent Signal**	
Cochlear Complex (27 genes)	Mammalian auditory hair cell	812–875 sp. (1, 676 aa)	4-mode AA+NT	10/27 sp. AA (18 SP)	+27.63 cum. (`USH2A`)	9/18 SP sonar ($3.02\times$, $p = 1.1 \times 10^{-3}$); 0 `ATG1`	Length-expected (0.77×); 50% SP bat/whale (`PCDH15`, `CDH23`, `LOXHD1`, `TRC1`)	
Myrmecophagous Guild	Pangolins ↔ anteaters/armadillos	882 sp. AA	20-state AA	4 (4 fam.)	+45.2 (`TTN`)	12.6× raw ($p = 3.3 \times 10^{-4}$); 1.35× len-adj.	Giant/multicopy loci (`TTN`, `ALMS1`, `OR10D3`, `ZSCAN16`; 1 non-mc $p = 0.157$)	
Subterranean & Marine Guilds	Blind moles/rodents; marine sp.	882 sp. AA	20-state AA	1 & 29	`ZDHHC21`; 21 fam.	$1.47\times$ ($p = 1.0$) & $1.31\times$ ($p = 0.164$, $q = 0.196$)	Proteome-wide cross-order rates match background expectation	
Bat–Whale & Echolocators	Echolocating mammals	882 sp. AA	20-state AA	28 / 59 cross-order	15 / 29 fam.	0.50× / 0.75× raw (1.05× fam. $p = 0.849$)	Proteome-wide neutral (1.01× `MALRUS`); localized in cochlear complex	
Tandem Paralog Arrays	Globin, CYP/UGT, OR, KRAB-ZNF	19, 953 AA	`INTERVAL`	2, 251 AA (1, 205 loci)	+507.3 max (med. 9.35, 85 aa)	88.3% intra-order (3.0–6.6× vs. 4.2% bg)	Globins (40.0%), CYP/UGT (32.8%), OR (26.3%), KRAB-ZNF (23.8%)	
SARS-CoV-2 Global + Blinded `X*`	Pandemic coronavirus	10, 062 × 29,903 bp	NT + Stage 4	3 + 6 `INT`	+52.61 / +31.41	165.4 s (≥ 28.8× lower bound vs. right-censored 4,758 s FT timeout); 1.18 s `X*`	Spike RBD [21,329, 22,917], ORF9–7a, `XBF`/`XBT`/`XAG`/`XBB`	
Global Dengue (`DENV-1`…`4`)	Urban arbovirus serotypes	5, 498 genomes	NT + AA	14 `INTERVAL`	+77.08 (`DENV-2`)	10.2–28.5 s (12.9–22.6× FT)	NS2A [3,632, 3,717], Capsid [8, 238] mosaic tracts	
`HIV-1` `KAL153` (`CRF03_AB`)	Circulating recombinant HIV	9 × 9,953 bp	NT + Stage 4	2 `INTERVAL`	+29.23	0.141 s (75, 955× `GARD`)	Exact `gag` [5, 2,558] and `nef` [9,102, 9,757] tracts	

Table S8: **Domain-wide prokaryotic marker analysis and quantitative test of the Complexity Hypothesis across** 242 **GTDB Release 95 markers (*24,25,88*).** Panel A reports single-gene nucleotide topological compatibility across 1,672 Archaea and 1,000 Bacteria (111 phyla). Panel B stratifies functional classes in nucleotide and native 20-state amino-acid space.

Panel A: Domain Stratum	Method	Res / Sp	Discordant FP	Prec.	RF (nRF)	Time (s)	Phylum Rec / Comp	Order Rec / Comp	Genus Rec / Comp
`bac120` Bacteria (120 genes, 111 phyla)	Bushwhacker (MDL)	335.6/857.8	192.6	42.7%	907.1 (0.5290)	23.03	13.8%/27.7%	28.0%/63.3%	67.6%/92.3%
	FastTree 2 (GTR + Γ)	857.6/857.8	495.5	42.2%	991.0 (0.5777)	20.36	29.6%/29.7%	59.0%/59.0%	86.0%/86.0%
`ar122` Archaea (122 genes, 19 phyla)	Bushwhacker (MDL)	855.2/1,387.4	449.6	51.1%	1,432.7 (0.4927)	16.04	22.4%/28.5%	38.0%/52.6%	68.1%/82.0%
	FastTree 2 (GTR + Γ)	1,387.2/1,387.4	624.1	55.0%	1,418.2 (0.4502)	25.20	33.4%/33.4%	55.0%/55.0%	81.6%/81.6%
Panel B: Complexity Hypothesis Class	**Markers (n)**	**Mean $\bar{L}$**	**Mean $\rho(A)$**	**BW Prec.**	**Inter-Phylum HGT/Gene**	**Total NT Jumps/Gene**	**`ar122` 20-State AA Shunt Loci**	**Cumulative AA Shunt Gain**	**Fisher Test**
Informational Core (Ribosome / RNAP)	85 NT / 20 AA	613 bp	31.2%	48.2%	13.81	25.08	0/20 (0.0%)	0.0 nats	OR = 0.0, $p = 0.213$, $q = 0.221$
Transcription & DNA Replication	43 NT	1,307 bp	25.6%	46.7%	14.14	24.74	0/1 (0.0%)	0.0 nats	Clade NT 0/10 vs. 3/58 ($p = 1.0$)
Operational Metabolic & Cellular	83 NT / 102 AA	938 bp	26.8%	45.9%	14.58	24.11	12/102 (11.8%, 15: 6 `INT`/9 `SP`)	249.6 nats	3 inter-phylum / 3 inter-class
Aminoacyl-tRNA Synthetases	31 NT	1,649 bp	26.0%	43.3%	15.29	27.13	Included in operational AA	`Aenigmat.`/`Micrarch.` ↔ `Nanoarch.`	DPANN exchanges

Table S9: **Lossless phylogenetic genome archiving (`.bwz`) compared against general-purpose dictionary compressors (`gzip -9`, `bzip2 -9`, `xz -6`/`lzma -9`) and mosaic recombination switchpoint titration.** Sizes are in bytes (verified bit-exact upon `bushwhacker -d` decompression); `Raw Seq. (B)` reports stripped header-free $N \times L$ ASCII sequence bytes fed identically to all general-purpose dictionary compressors (`gzip -9`, `bzip2 -9` including column-transposed layout, and `xz -6`/`lzma -9`), alongside `.bwz` sequence-payload bytes (`Payload`) and full self-contained `.bwz` archive bytes including taxon names, Newick tree, and headers (`Archive`), with Ratio > 1.00× indicating `.bwz` outperforms the best general-purpose compressor and Ratio < 1.00× flagging genome-wide mosaicism.

Dataset / Regime	Category	$N \times L$	Raw Seq. (B)	`gzip -9`	`bzip2 -9`	`xz -6` / `lzma -9`	`.bwz` Payload (B)	`.bwz` Archive (B)	Payload Ratio vs. Best GP (`gzip` ratio)
`SARS-CoV-2_10062`	Pandemic Viral	10,062 × 29,903	300,883,986	7,322,102	2,641,139	797,044	387,697 (38.5 B/g)	635,851 (63.2 B/g)	2.06× (18.89× gz; arch. 1.25×)
`SARS-CoV-2_5097`	Pandemic Viral	5,097 × 29,903	152,415,591	3,791,027	1,312,281	409,108	220,588 (43.3 B/g)	340,571 (66.8 B/g)	1.86× (17.19× gz; arch. 1.20×)
`DENV-1..4` (4 cohorts)	Pandemic Arboviral	5,498 × 10,050–10,065	55,315,569	1,735,762	—	415,904	—	295,642	Arch. 1.26–1.52× xz (5.23–6.42× gz)
`InfluenzaA.fasta`	Seasonal Viral	349 × 987	344,463	6,729	4,813	5,005	4,133	10,371	1.17× (1.63× gz; arch. 0.46×)
`Human_mtDNA`	Organelle Genome	78 × 16,585	1,293,630	23,424	9,709	9,348	6,054	8,345	1.54× (3.87× gz; arch. 1.12×)
`prestin.fasta`	Mammalian Gene	116 × 2,235	259,260	27,291	19,154	11,821	7,503	10,554	1.58× (3.64× gz; arch. 1.12×)
`rbcL_483`	Deep Plastid	483 × 1,398	675,234	54,956	44,819	31,835	22,061	33,437	1.44× (2.49× gz; arch. 0.95×)
`16S_N2048`	Microbiome 16S	2,048 × 1,348	2,760,704	406,975	277,753	148,878	106,817	156,355	1.39× (3.81× gz; arch. 0.95×)
`16S_N32768`	Microbiome 16S	32,768 × 1,348	44,171,264	6,304,051	—	2,284,260	—	2,111,286 (64.4 B/OTU)	Arch. 1.08× xz (2.99× gz)
`PMEN1` (24.4 Mbp)	Bacterial Chrom.	11 × 2,221,315	24,434,465	6,434,278	712,233	600,352	594,506	594,774	1.01× (10.82× gz; arch. 1.01×)
`Hominoidea_MHC` / `pBP2x`	Local Conv. / Mosaic	379–12	427,512/248,196	9,944/11,707	7,438/8,131	6,402/7,144	4,180/5,928	12,664/6,169	1.53× / 1.21× (arch. 0.51× / 1.16×)
`Sarbeco_B3` (31 kb)	Whole-Genome Mosaic	83 × 30,965	2,570,095	222,223	103,100	78,441	110,499	114,718	0.71× (arch. 0.68×; Mosaic)
`HIV1_CRF02_AG`	Whole-Genome Mosaic	9 × 9,953	89,577	16,121	10,798	9,524	14,768	15,018	0.65× (arch. 0.63×; Mosaic)
Titration $K = 0 \to 1 \to 8 \to 32$	0–32 bkpts/mosaic	56 × 12,000	672,000	168.4–180.4k	94.4–102.6k	54.4 → 59.5 → 55.8 → 57.4k	43.7 → 47.3 → 50.0 → 59.6k	45.2 → 48.7 → 51.9 → 61.9k	1.24 → 1.26 → 1.12 → 0.96× (arch. 1.20 → 0.93×)

Table S10: **Zero-leakage single-cell human tumor phylogenomics across** 1,001,609 **myeloid cells (ordered 3-state somatic zygosity) and** 4,950 **solid-tumor single-cell whole-genome copy-number profiles (ordered 12-state CNA).** All cell counts, anatomic labels, and passage numbers were withheld during tree and network inference; DLP+ `Sitka` comparisons used author-published phylogenies without re-executing the `Sitka` JVM/Nextflow pipeline.

Cohort / Modality	Representation (`-ordered`)	Cells / Runs	Features (L)	Resolved / Max Splits	Collapsed%	Compression	Runtime (s)	Post-Hoc Biological Validation
COMPASS Myeloid scDNA (*47*)	Unthresholded 3-state zygosity	1,001,609 (168 runs)	3–28 driver loci	6,222 non-WT clones	Burst-calibrated	—	2.62 s	$k \geq 3$ hubs 6.18× cells (run $p = 1.7 \times 10^{-18}$; depth $p = 5.9 \times 10^{-12}$; FE $p = 1.4 \times 10^{-28}$)
Morita AML Tapestri (*27*)	Curated + raw scDNA	123 pts (465 clones)	31 / 16 loci	100 branching pts	`AML-38-001` 3.0–4.1 ms vs. `COMPASS` 471.4 s (3.54 GB)		0.128 s	FE OLS $p = 3.9 \times 10^{-6}$; late signaling OR = 1.79 ($p = 1.9 \times 10^{-3}$); 44 pts parallel
`OV2295` HGSOC (*48*)	Colinear segments / Raw 500-kb bins	1,111 cells	420 segs / 4,375 bins	631/1,108 (648 bin)	43.1% (41.5%)	6.0× / 29.4×	4.66 s / 20.75 s	Ovary vs. ascites $F_1 = 0.997/0.995$ (95.7% pur.); $\rho = 0.818$
`SA501` TNBC PDX (*48*)	Colinear segments / Raw 500-kb bins	2,691 cells	365 segs / 4,375 bins	1,639/2,688 (1,799 bin)	39.0% (33.1%)	6.5× / 33.5×	11.30 s / 65.53 s	Multi-passage sweeps (`X2` → `X15`); 60.7% bin purity
`SA609` TNBC PDX (*48*)	Colinear segments / Raw 500-kb bins	600 cells	448 segs / 4,375 bins	280/597 (299 bin)	53.1% (49.9%)	4.2× / 21.9×	4.68 s / 19.97 s	Diploid normal (`SA928`, 91.3% dominant clade) vs. `X2`
`SA535` TNBC PDX (*48*)	Colinear segments / Raw 500-kb bins	548 cells	426 segs / 4,375 bins	263/545 (285 bin)	51.7% (47.7%)	6.0× / 29.8×	1.26 s / 7.35 s	Subclonal passage clades (91.5% pur.; 21.9 s seg; 113.6 s bin)

Table S11: **Continuous-time reversible hard-CAT rate-category compression (`GTR+CAT8` and `LG08+CAT8`) with 64-site cache-tiled pruning and four-mode Eq.** (9) **ablations (all ToL benchmarks on 32 CPU cores).** Panel A scores tree precision/recall/nRF and rank monophyly against published TOGA2 (882-mammal) and GTDB r95 Archaeal/Bacterial reference phylogenies; in `bac120` (1,000 genomes spanning all 111 GTDB phyla), 6 singleton phyla lack internal splits, leaving 105 non-singleton phyla in the Rank 1 monophyly denominator.

Panel A: Tree-of-Life Supermatrix	Method (32 Threads)	Runtime (s)	Speedup	Peak RAM (GB)	Prec. / Rec. (nRF)	Rank 1 (Superord./Phylum)	Rank 2 (Order/Class)	Rank 3 (Family/Order)	Rank 4 (Genus/Family)
Mammalia Coding DNA	**Bushwhacker (`GTR+CAT8` MDL)**	**31.2–48.6**	**1.0×**	**2.38**	**91.9**%/**91.8**% (**0.081**)	**100.0**% (4/4)	**100.0**% (18/18)	**95.5**% (84/88)	**94.0**% (172/183)
(882 sp. ×543,864 bp,	IQ-TREE 2 (`GTR+G4`, fast)	1,166.56	24.0× slower	39.82 (16.7×)	92.4%/92.4% (0.076)	100.0%	100.0%	95.5%	94.5%
253 single-copy loci; 92.7% BW–IQ)	FastTreeMP (`GTR+CAT`)	13,990.59	287.9× slower	—	92.6%/92.6% (0.074)	100.0%	100.0%	95.5%	94.5%
Mammalia Protein	**Bushwhacker (`LG08+CAT8` MDL)**	**49.0–65.8**	**1.0×**	**1.98**	**95.8**%/**83.6**% (**0.100**; 32 FP)	**100.0**% (4/4)	**100.0**% (18/18)	**95.5**% (84/88)	**92.4**% (169/183)
(882 sp. ×181,288 AA,	IQ-TREE 2 (`LG+G4`, fast)	2,434.49	37.0× slower	79.15 (40.0×)	91.6%/91.6% (0.084; 74 FP)	100.0%	100.0%	95.5%	93.4%
112 silent AA stems contracted)	FastTreeMP (`LG+CAT`)	12,711.41	193.2× slower	—	91.4%/91.4% (0.086; 76 FP)	100.0%	100.0%	95.5%	93.4%
Archaea Domain (`ar122`)	**Bushwhacker (`LG08+CAT8` MDL)**	**26.7–31.5**	**1.0×**	**2.57**	**84.4**%/**84.4**% (**0.156**)	**82.4**% (14/17)	**90.0**% (36/40)	**98.8**% (84/85)	**98.6**% (146/148)
(1,672 sp. ×32,675 AA,	IQ-TREE 2 (`LG+G4`, fast)	1,691.53	53.7× slower	32.95 (12.8×)	86.9%/86.9% (0.131)	88.2%	95.0%	98.8%	100.0%
19 phyla, ~ 3.5 Gyr)	FastTreeMP (`LG+CAT`)	4,504.04	143.0× slower	10.21 (4.0×)	86.5%/86.5% (0.135)	88.2%	92.5%	97.7%	99.3%
	Unweighted-Fitch Parsimony Ablation	202.32	6.4× slower	—	37.5%/37.2% (0.624)	5.9% (1/17)	7.5%	20.0%	28.4%
Bacteria Domain (`bac120`)	**Bushwhacker (`LG08+CAT8` MDL)**	**24.3–26.9**	**1.0×**	**2.71**	**75.3**%/**75.3**% (**0.247**)	**94.3**% (99/105)	**97.1**% (166/171)	**99.6**% (222/223)	**99.0**% (203/205)
(1,000 sp. ×43,733 AA,	IQ-TREE 2 (`LG+G4`, fast)	1,137.85	42.3× slower	26.37 (9.7×)	76.4%/76.4% (0.236)	94.3%	98.3%	99.6%	98.5%
all 111 phyla, ~ 3.5 Gyr)	FastTreeMP (`LG+CAT`)	3,822.63	142.1× slower	—	77.2%/77.2% (0.228)	89.5%	95.3%	99.6%	99.5%
	Unweighted-Fitch Parsimony Ablation	291.21	10.8× slower	—	26.6%/26.4% (0.733)	15.2% (16/105)	30.4%	39.9%	43.9%
Panel B: Eq. (9) Four-Mode Ablation	Configuration / Baseline	`S08b` FP	`S10b` FP	`P2` FP	`sim_bic` FP ($n = 500$)	Mammalia AA nRF	`ar122` / `bac120` nRF	`Anomaly-24` nRF	`Anomaly-24` Wall / CPU
Fixed $\Gamma(1,1)$ + Mult. Penalty	**Default Bushwhacker**	0.15	0.05	0.04	0.17	0.1001	0.1558/0.2467	0.3000	0.60 s / 6.23 s
Drop $\frac{1}{2}\ln(N-3)$ Term	`-no-mult-pen`	0.60	0.40	0.12	0.22	0.1001	0.1558/0.2467	0.2857	0.58 s / 6.19 s
Per-Gene α Rate Fit	`-empirical-alpha`	0.15	0.05	0.04	0.20	0.1115	0.1576/0.2357	0.3286	0.58 s / 6.17 s
Drop Mult. + Empirical α	`-no-mult-pen -empirical-alpha`	0.60	0.35	0.12	0.25	0.1121	0.1576/0.2357	0.2905	0.56 s / 6.14 s
Support-Collapsed ML Baseline	IQ-TREE 2 (`UFBoot95` / `SH-aLRT80`)	—	—	—	—	—	—	0.3238/0.1905	37.5 s / 2,372.0 s

References and Notes

1. W. M. Fitch, Toward defining the course of evolution: minimum change for a specific tree topology. *Systematic Zoology* **20** (4), 406–416 (1971), doi:10.2307/2412116.

2. J. Felsenstein, Evolutionary trees from DNA sequences: a maximum likelihood approach. *Journal of Molecular Evolution* **17** (6), 368–376 (1981), doi:10.1007/BF01734359.

3. N. Saitou, M. Nei, The neighbor-joining method: a new method for reconstructing phylogenetic trees. *Molecular Biology and Evolution* **4** (4), 406–425 (1987), doi:10.1093/oxfordjournals.molbev.a040454.

4. M. N. Price, P. S. Dehal, A. P. Arkin, FastTree 2 – approximately maximum-likelihood trees for large alignments. *PLoS ONE* **5** (3), e9490 (2010), doi:10.1371/journal.pone.0009490.

5. A. M. Kozlov, D. Darriba, T. Flouri, B. Morel, A. Stamatakis, RAxML-NG: a fast, scalable and user-friendly tool for maximum likelihood phylogenetic inference. *Bioinformatics* **35** (21), 4453–4455 (2019), doi:10.1093/bioinformatics/btz305.

6. B. Q. Minh, *et al.*, IQ-TREE 2: new models and efficient methods for phylogenetic inference in the genomic era. *Molecular Biology and Evolution* **37** (5), 1530–1534 (2020), doi:10.1093/molbev/msaa015.

7. J. Rissanen, Modeling by shortest data description. *Automatica* **14** (5), 465–471 (1978), doi:10.1016/0005-1098(78)90005-5.

8. J. Rissanen, A universal prior for integers and estimation by minimum description length. *The Annals of Statistics* **11** (2), 416–431 (1983), doi:10.1214/aos/1176346150.

9. J. Felsenstein, Confidence limits on phylogenies: an approach using the bootstrap. *Evolution* **39** (4), 783–791 (1985), doi:10.1111/j.1558-5646.1985.tb00420.x.

10. S. L. Kosakovsky Pond, D. Posada, M. B. Gravenor, C. H. Woelk, S. D. W. Frost, Automated phylogenetic detection of recombination using a genetic algorithm. *Molecular Biology and Evolution* **23** (10), 1891–1901 (2006), doi:10.1093/molbev/msl051.

11. A. Rokas, S. B. Carroll, Bushes in the tree of life. *PLoS Biology* **4** (11), e352 (2006), doi:10.1371/journal.pbio.0040352.

12. X.-X. Shen, C. T. Hittinger, A. Rokas, Contentious relationships in phylogenomic studies can be driven by a handful of genes. *Nature Ecology & Evolution* **1** (5), 0126 (2017), doi:10.1038/s41559-017-0126.

13. S. Abadi, D. Azouri, T. Pupko, I. Mayrose, Model selection may not be a mandatory step for phylogeny reconstruction. *Nature Communications* **10** (1), 934 (2019), doi:10.1038/s41467-019-08822-w.

14. J. Felsenstein, Cases in which parsimony or compatibility methods will be positively misleading. *Systematic Zoology* **27** (4), 401–410 (1978), doi:10.2307/2412923.

15. Z. Yang, Maximum likelihood phylogenetic estimation from DNA sequences with variable rates over sites: approximate methods. *Journal of Molecular Evolution* **39** (3), 306–314 (1994), doi: 10.1007/BF00160154.

16. S. Q. Le, O. Gascuel, An improved general amino acid replacement matrix. *Molecular Biology and Evolution* **25** (7), 1307–1320 (2008), doi:10.1093/molbev/msn067.

17. W. F. Doolittle, Phylogenetic classification and the universal tree. *Science* **284** (5423), 2124–2128 (1999), doi:10.1126/science.284.5423.2124.

18. M. Worobey, A. Rambaut, E. C. Holmes, Widespread intra-serotype recombination in natural populations of dengue virus. *Proceedings of the National Academy of Sciences* **96** (13), 7352–7357 (1999), doi:10.1073/pnas.96.13.7352.

19. Y. Liu, *et al.*, Convergent sequence evolution between echolocating bats and dolphins. *Current Biology* **20** (2), R53–R54 (2010), doi:10.1016/j.cub.2009.11.058.

20. J. Parker, *et al.*, Genome-wide signatures of convergent evolution in echolocating mammals. *Nature* **502** (7470), 228–231 (2013), doi:10.1038/nature12511.

21. Z. Zou, J. Zhang, No genome-wide protein sequence convergence for echolocation. *Molecular Biology and Evolution* **32** (5), 1237–1241 (2015), doi:10.1093/molbev/msv014.

22. Y. Turakhia, *et al.*, Pandemic-scale phylogenomics reveals the SARS-CoV-2 recombination landscape. *Nature* **609** (7929), 994–997 (2022), doi:10.1038/s41586-022-05189-9.

23. Y. V. Malovichko, *et al.*, Accurate, comprehensive gene annotation and ortholog identification across thousands of vertebrate genomes with TOGA2. *bioRxiv* (2026), doi:10.64898/2026.06.30.735536.

24. R. Jain, M. C. Rivera, J. A. Lake, Horizontal gene transfer among genomes: the complexity hypothesis. *Proceedings of the National Academy of Sciences* **96** (7), 3801–3806 (1999), doi: 10.1073/pnas.96.7.3801.

25. D. H. Parks, *et al.*, A complete domain-to-species taxonomy for Bacteria and Archaea. *Nature Biotechnology* **38** (9), 1079–1086 (2020), doi:10.1038/s41587-020-0501-8.

26. W. S. DeWitt, *et al.*, Replaying germinal center evolution on a quantified affinity landscape. *Cell* (2026).

27. K. Morita, *et al.*, Clonal evolution of acute myeloid leukemia revealed by high-throughput single-cell genomics. *Nature Communications* **11** (1), 5327 (2020), doi:10.1038/s41467-020-19119-8.

28. Materials and methods are available as supplementary materials.

29. C. Zhang, R. Nielsen, S. Mirarab, ASTER: A package for large-scale phylogenomic reconstructions. *Molecular Biology and Evolution* **42** (8), msaf172 (2025), doi:10.1093/molbev/msaf172.

30. E. D. Jarvis, *et al.*, Whole-genome analyses resolve early branches in the tree of life of modern birds. *Science* **346** (6215), 1320–1331 (2014), doi:10.1126/science.1253451.

31. R. O. Prum, *et al.*, A comprehensive phylogeny of birds (Aves) using targeted next-generation DNA sequencing. *Nature* **526** (7574), 569–573 (2015), doi:10.1038/nature15697.

32. N. J. Wickett, *et al.*, Phylotranscriptomic analysis of the origin and early diversification of land plants. *Proceedings of the National Academy of Sciences* **111** (45), E4859–E4868 (2014), doi:10.1073/pnas.1323926111.

33. B. Misof, *et al.*, Phylogenomics resolves the timing and pattern of insect evolution. *Science* **346** (6210), 763–767 (2014), doi:10.1126/science.1257570.

34. M.-Y. Chen, D. Liang, P. Zhang, Selecting question-specific genes to reduce incongruence in phylogenomics: a case study of jawed vertebrate backbone phylogeny. *Systematic Biology* **64** (6), 1104–1120 (2015), doi:10.1093/sysbio/syv059.

35. M. L. Borowiec, E. K. Lee, J. C. Chiu, D. C. Plachetzki, Extracting phylogenetic signal and accounting for bias in whole-genome data sets supports the Ctenophora as sister to remaining Metazoa. *BMC Genomics* **16**, 987 (2015), doi:10.1186/s12864-015-2146-4.

36. N. V. Whelan, *et al.*, Ctenophore relationships and their placement as the sister group to all other animals. *Nature Ecology & Evolution* **1** (11), 1737–1746 (2017), doi:10.1038/s41559-017-0331-3.

37. S. Song, L. Liu, S. V. Edwards, S. Wu, Resolving conflict in eutherian mammal phylogeny using phylogenomics and the multispecies coalescent model. *Proceedings of the National Academy of Sciences* **109** (37), 14942–14947 (2012), doi:10.1073/pnas.1211733109.

38. A. J. Shultz, T. B. Sackton, Immune genes are hotspots of shared positive selection across birds and mammals. *eLife* **8**, e41815 (2019), doi:10.7554/eLife.41815.

39. G. W. C. Thomas, M. W. Hahn, Determining the null model for detecting adaptive convergence from genomic data: a case study using echolocating mammals. *Molecular Biology and Evolution* **32** (5), 1232–1236 (2015), doi:10.1093/molbev/msv013.

40. P.-A. Christin, *et al.*, Adaptive evolution of C_4 photosynthesis through recurrent lateral gene transfer. *Current Biology* **22** (5), 445–449 (2012), doi:10.1016/j.cub.2012.01.054.

41. S. Lytras, *et al.*, Exploring the natural origins of SARS-CoV-2 in the light of recombination. *Genome Biology and Evolution* **14** (2), evac018 (2022), doi:10.1093/gbe/evac018.

42. K. L. Wyres, M. M. C. Lam, K. E. Holt, Population genomics of *Klebsiella pneumoniae*. *Nature Reviews Microbiology* **18** (6), 344–359 (2020), doi:10.1038/s41579-019-0315-1.

43. C. J. Castelle, J. F. Banfield, Major new microbial groups expand diversity and alter our understanding of the tree of life. *Cell* **172** (6), 1181–1197 (2018), doi:10.1016/j.cell.2018.02.016.

44. D. McDonald, *et al.*, An improved Greengenes taxonomy with explicit ranks for ecological and evolutionary analyses of bacteria and archaea. *The ISME Journal* **6** (3), 610–618 (2012), doi:10.1038/ismej.2011.139.

45. J. M. J. Tas, *et al.*, Visualizing antibody affinity maturation in germinal centers. *Science* **351** (6277), 1048–1054 (2016), doi:10.1126/science.aad3439.

46. W. S. DeWitt, L. Mesin, G. D. Victora, V. N. Minin, F. A. Matsen, IV, Using genotype abundance to improve phylogenetic inference. *Molecular Biology and Evolution* **35** (5), 1253–1265 (2018), doi:10.1093/molbev/msy020.

47. E. Sollier, J. Kuipers, K. Takahashi, N. Beerenwinkel, K. Jahn, COMPASS: joint copy number and mutation phylogeny reconstruction from amplicon single-cell sequencing data. *Nature Communications* **14** (1), 4921 (2023), doi:10.1038/s41467-023-40378-8.

48. S. Salehi, *et al.*, Cancer phylogenetic tree inference at scale from 1000s of single cell genomes. *Peer Community Journal* **3**, e63 (2023), doi:10.24072/pcjournal.292.

49. S. L. Kosakovsky Pond, *et al.*, Bushwhacker: Evolutionary history from genomes to single cells by lossless compression, version 1.0, GitHub and Zenodo (2026), doi:10.5281/zenodo.24222789, `https://github.com/veg/bushwhacker`.

50. H. Akaike, A new look at the statistical model identification. *IEEE Transactions on Automatic Control* **19** (6), 716–723 (1974), doi:10.1109/TAC.1974.1100705.

51. G. Schwarz, Estimating the dimension of a model. *The Annals of Statistics* **6** (2), 461–464 (1978), doi:10.1214/aos/1176344136.

52. D. Posada, T. R. Buckley, Model selection and model averaging in phylogenetics: advantages of Akaike information criterion and Bayesian approaches over likelihood ratio tests. *Systematic Biology* **53** (5), 793–808 (2004), doi:10.1080/10635150490522304.

53. S. Kalyaanamoorthy, B. Q. Minh, T. K. F. Wong, A. von Haeseler, L. S. Jermiin, ModelFinder: fast model selection for accurate phylogenetic estimates. *Nature Methods* **14** (6), 587–589 (2017), doi:10.1038/nmeth.4285.

54. G. E. P. Box, Science and statistics. *Journal of the American Statistical Association* **71** (356), 791–799 (1976), doi:10.1080/01621459.1976.10480949.

55. E. Susko, A. J. Roger, On the use of information criteria for model selection in phylogenetics. *Molecular Biology and Evolution* **37** (2), 549–562 (2020), doi:10.1093/molbev/msz228.

56. M. Anisimova, M. Gil, J.-F. Dufayard, C. Dessimoz, O. Gascuel, Survey of branch support methods demonstrates accuracy, power, and robustness of fast likelihood-based approximation schemes. *Systematic Biology* **60** (5), 685–699 (2011), doi:10.1093/sysbio/syr041.

57. D. T. Hoang, O. Chernomor, A. von Haeseler, B. Q. Minh, L. S. Vinh, UFBoot2: improving the ultrafast bootstrap approximation. *Molecular Biology and Evolution* **35** (2), 518–522 (2018), doi:10.1093/molbev/msx281.

58. L. Jin, M. Nei, Limitations of the evolutionary parsimony method of phylogenetic analysis. *Molecular Biology and Evolution* **7** (1), 82–102 (1990), doi:10.1093/oxfordjournals.molbev.a040588.

59. S. Guindon, *et al.*, New algorithms and methods to estimate maximum-likelihood phylogenies: assessing the performance of PhyML 3.0. *Systematic Biology* **59** (3), 307–321 (2010), doi:10.1093/sysbio/syq010.

60. M. O. Dayhoff, R. M. Schwartz, B. C. Orcutt, A model of evolutionary change in proteins, in *Atlas of Protein Sequence and Structure*, M. O. Dayhoff, Ed. (National Biomedical Research Foundation, Washington, DC), vol. 5 of *Supplement 3*, pp. 345–352 (1978).

61. S. Tavaré, Some probabilistic and statistical problems in the analysis of DNA sequences, in *Lectures on Mathematics in the Life Sciences* (American Mathematical Society, Providence, RI), vol. 17, pp. 57–86 (1986).

62. I. H. Witten, R. M. Neal, J. G. Cleary, Arithmetic coding for data compression. *Communications of the ACM* **30** (6), 520–540 (1987), doi:10.1145/214762.214771.

63. K. Jahn, J. Kuipers, N. Beerenwinkel, Tree inference for single-cell data. *Genome Biology* **17**, 86 (2016), doi:10.1186/s13059-016-0936-x.

64. D. F. Robinson, L. R. Foulds, Comparison of phylogenetic trees. *Mathematical Biosciences* **53** (1-2), 131–147 (1981), doi:10.1016/0025-5564(81)90043-2.

65. M. Gerth, Neglecting model selection alters phylogenetic inference. *bioRxiv* p. 849018 (2019), doi:10.1101/849018.

66. X. Zhou, X.-X. Shen, C. T. Hittinger, A. Rokas, Evaluating fast maximum likelihood-based phylogenetic programs using empirical phylogenomic data sets. *Molecular Biology and Evolution* **35** (2), 486–503 (2018), doi:10.1093/molbev/msx302.

67. Y. Li, Z. Liu, P. Shi, J. Zhang, The hearing gene Prestin unites echolocating bats and whales. *Current Biology* **20** (2), R55–R56 (2010), doi:10.1016/j.cub.2009.11.042.

68. B. M. Kirilenko, *et al.*, Integrating gene annotation with orthology inference at scale. *Science* **380** (6643), eabn3107 (2023), doi:10.1126/science.abn3107.

69. Z. Xi, L. Liu, J. S. Rest, C. C. Davis, Coalescent versus concatenation methods and the placement of *Amborella* as sister to water lilies. *Systematic Biology* **63** (6), 919–932 (2014), doi:10.1093/sysbio/syu055.

70. P.-A. Christin, N. Salamin, V. Savolainen, M. R. Duvall, G. Besnard, C_4 photosynthesis evolved in grasses via parallel adaptive genetic changes. *Current Biology* **17** (14), 1241–1247 (2007), doi:10.1016/j.cub.2007.06.036.

71. L. T. Dunning, *et al.*, Lateral transfers of large DNA fragments spread functional genes among grasses. *Proceedings of the National Academy of Sciences* **116** (10), 4416–4425 (2019), doi:10.1073/pnas.1810031116.

72. D. H. Parks, *et al.*, A standardized bacterial taxonomy based on genome phylogeny substantially revises the tree of life. *Nature Biotechnology* **36** (10), 996–1004 (2018), doi:10.1038/nbt.4229.

73. N. J. Croucher, *et al.*, Rapid pneumococcal evolution in response to clinical interventions. *Science* **331** (6016), 430–434 (2011), doi:10.1126/science.1198545.

74. J. K. Carr, *et al.*, Full genome sequences of human immunodeficiency virus type 1 subtypes G and A/G intersubtype recombinants. *Virology* **247** (1), 22–31 (1998), doi:10.1006/viro.1998.9211.

75. D. L. Robertson, *et al.*, HIV-1 nomenclature proposal. *Science* **288** (5463), 55–56 (2000), doi: 10.1126/science.288.5463.55d.

76. M. F. Boni, *et al.*, Evolutionary origins of the SARS-CoV-2 sarbecovirus lineage responsible for the COVID-19 pandemic. *Nature Microbiology* **5** (11), 1408–1417 (2020), doi: 10.1038/s41564-020-0771-4.

77. P. Parham, T. Ohta, Population biology of antigen presentation by MHC class I molecules. *Science* **272** (5258), 67–74 (1996), doi:10.1126/science.272.5258.67.

78. C. Pardo-Diaz, *et al.*, Adaptive introgression across species boundaries in *Heliconius* butterflies. *PLoS Genetics* **8** (6), e1002752 (2012), doi:10.1371/journal.pgen.1002752.

79. Heliconius Genome Consortium, Butterfly genome reveals promiscuous exchange of mimicry adaptations among species. *Nature* **487** (7405), 94–98 (2012), doi:10.1038/nature11041.

80. N. Ly-Trong, S. Naser-Khdour, R. Lanfear, B. Q. Minh, AliSim: a fast and versatile phylogenetic sequence simulator for the genomic era. *Molecular Biology and Evolution* **39** (5), msac092 (2022), doi:10.1093/molbev/msac092.

81. J. H. Degnan, N. A. Rosenberg, Discordance of species trees with their most likely gene trees. *PLoS Genetics* **2** (5), e68 (2006), doi:10.1371/journal.pgen.0020068.

82. J. Ripplinger, J. Sullivan, Does choice in model selection affect maximum likelihood analysis? *Systematic Biology* **57** (1), 76–85 (2008), doi:10.1080/10635150801898920.

83. S. J. Spielman, Relative model fit does not predict topological accuracy in single-gene protein phylogenetics. *Molecular Biology and Evolution* **37** (7), 2110–2123 (2020), doi:10.1093/molbev/msaa075.

84. J. E. Tarver, *et al.*, The interdependence of molecular clock models and tree topologies in the estimation of placental mammal divergence times. *Genome Biology and Evolution* **8** (2), 330–344 (2016), doi:10.1093/gbe/evv261.

85. C. Zhang, M. Rabiee, E. Sayyari, S. Mirarab, ASTRAL-III: polynomial time species tree reconstruction from partially resolved gene trees. *BMC Bioinformatics* **19** (Suppl 6), 153 (2018), doi:10.1186/s12859-018-2129-y.

86. K. T. J. Davies, J. A. Cotton, J. D. Kirwan, E. C. Teeling, S. J. Rossiter, Parallel signatures of sequence evolution among hearing genes in echolocating mammals: an emerging model of genetic convergence. *Heredity* **108** (5), 480–489 (2012), doi:10.1038/hdy.2011.119.

87. Y.-Y. Shen, L. Liang, G.-S. Li, R. W. Murphy, Y.-P. Zhang, Parallel evolution of auditory genes for echolocation in bats and toothed whales. *PLoS Genetics* **8** (6), e1002788 (2012), doi:10.1371/journal.pgen.1002788.

88. C. R. Woese, G. J. Olsen, M. Ibba, D. Söll, Aminoacyl-tRNA synthetases, the genetic code, and the evolutionary process. *Microbiology and Molecular Biology Reviews* **64** (1), 202–236 (2000), doi:10.1128/MMBR.64.1.202-236.2000.

89. C. G. Dowson, *et al.*, Horizontal transfer of penicillin-binding protein genes in penicillin-resistant clinical isolates of *Streptococcus pneumoniae*. *Proceedings of the National Academy of Sciences* **86** (22), 8842–8846 (1989), doi:10.1073/pnas.86.22.8842.

90. N. Li, M. Stephens, Modeling linkage disequilibrium and identifying recombination hotspots using single-nucleotide polymorphism data. *Genetics* **165** (4), 2213–2233 (2003), doi:10.1093/genetics/165.4.2213.

91. O. Gascuel, BIONJ: an improved version of the NJ algorithm based on a simple model of sequence data. *Molecular Biology and Evolution* **14** (7), 685–695 (1997), doi:10.1093/oxfordjournals.molbev.a025808.